Class 6
Chemistry

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1. Nature of Matter

Solutions

Classification of Matter

LEVEL-I

1. Correct option: (C)

Anything that has some mass and occupies space is called matter. The quantity of matter present in an object is mass.

2. Correct option: (D)

Sun, the rays of the sun emit energy in the form radiation.

3. Correct option: (A)

Solids are rigid having definite shape and volume

4. Correct option: (C)

Solids, liquids and gases are called the three states of matter.

5. Correct option: (A)

In three states of matter, gases have neither definite shape nor definite volume.

6. Correct option: (C)

The maximum intermolecular force of attraction observed in solids because of closely packing.

7. Correct option: (B)

The liquid molecules can move freely because, the molecules are packed loosely.

8. Correct option: (A)

A) Solids have definite volume and shape.

B) Liquids have definite volume and indefinite shape.

C) Intermolecular force of attraction in solids is high.

9. Correct option: (B)

The liquid state of matter has definite volume and no definite shape.

10. Correct option: (B)

The gas has negligible intermolecular force of attraction because of very loose packing of molecules.

11. Correct option: (C)

Gas has the property of flowing in all directions. Because of that we smell the fragrance of Perfumes, flowers etc.

12. Correct option: (C)

A) Liquids take the shape of vessel in which it present. So, it does not have definite shape.

B) Liquids have definite volume.

C) Liquids can be compressed slightly because of loosely packing and intermolecular space.

D) Liquids have more intermolecular force of attraction compared to gases because of close packing.

13. Correct option: (A)

Solid state of matter does not need a vessel to contain them because of definite shape and definite volume.

14. Correct option: (D)

The existence of matter in different states depends on different factors, but primarily depends on both temperature and pressure. These two factors influence

the arrangement and motion of atoms or molecules of different substances that made up of, leading to the different states of matter.

15. Correct option: (C)
Gas has the negligible intermolecular force because the molecules in gas are very loosely packed.

16. Correct option: (D)
When small amount of sugar is dissolved in it, the volume of the solution will not change (minimal change), because the sugar molecules occupy the space between the water molecules and the sugar molecules are surrounded by the molecules of water.

17. Correct option: (D)
Inter particle spaces are minimum in solids not in gaseous state of the substance.

18. Correct option: (C)
Because of very loosely packing and more intermolecular space gases have more compressibility.

19. Correct option: (C)
Gases have more intermolecular space because of very loose packing.

20. Correct option: (C)
The molecules in gases are very loosely packed.

21. Correct option: (A)
The molecules in solids are closely packed.

22. Correct option: (D)
Iron bar has the maximum intermolecular force of attraction, because iron bar exists in solid state where as the remaining substances exist in liquid and gases state which have less inter molecular force of attraction compared to solid state substances.

23. Correct option: (A)
Iodine exists as solid; Bromine exists as liquid and fluorine exists as gas.

24. Correct option: (A)
As the molecules in solids are closely packed, solids cannot be compressed.

25. Correct option: (A)
The molecules in solid state are closely packed, they cannot move freely. So, collisions are less in solid state.

26. Correct option: (C)
Gas state of matter shows diffusion property because of very loose packing of these molecules they can move freely from one place to another place.

27. Correct option: (C)
Gas state of matter does not settle at the bottom of the container because of diffusion property.

28. Correct option: (D)
Because of high temperature the matter exists as plasma in sun and core of stores.

29. Correct option: (D)
Plasma consists of ionization of atoms or molecules. For ionization high temperature required so plasma state of matter exist at very high temperature.

30. Correct option: (D)
Both temperature and pressure are the main factors effecting the inter conversion of the states of matter.

LEVEL-II

1. Correct option: (B)

A) Intermolecular force of attraction is more in solids.

B) Intermolecular force of attraction in gases is minimum.

C) Intermolecular space in solids is minimum.

D) Intermolecular space is present in liquids.

2. Correct option: (B)

Feeling of cold is not a matter, which does not contain both volume and mass.

3. Correct option: (B)

Compared to liquids gases are more compressible.

4. Correct option: (D)

Gases present in atmosphere are not application of high compressibility of gases. Whereas the remaining all are stored by compressing.

5. Correct option: (D)

The properties we exploited in the usage of perfume are diffusion property and expansibility of gases, because of that perfume move from one place to another place and we can sense the fragrance of perfume.

6. Correct option: (B)

In the given substances, Iodine exists as solid at room temperature and the remaining substances exist as liquids and gases. So, iodine has strong force of attraction.

7. Correct option: (A)

Because of very loose packing and more intermolecular force of attraction gases diffuse easily and forms homogeneous mixture due to this property.

8. Correct option: (D)

The substances which exist as solids at room temperature will have strong intermolecular forces of attraction than those exist as liquids and gases. So, carbon and potassium have strong intermolecular forces of attraction as they exist as solids at room temperature.

9. Correct option: (B)

Because of close packing of molecules in solids, the intermolecular space is less and the molecules cannot move freely and the kinetic energy is less compared to liquid and gaseous state. So thermal expansion is less in solid state.

10. Correct option: (D)

Bose Einstein condensate is a condensate super fluid phase formed by atoms, cooled very near to absolute zero temperature.

11. Correct options: (A, B, C, D)

Gases are highly compressible, highly expansible, exert pressure in all directions and have low density. These are properties of gases.

12. Correct options: (A, B, C, D)

Matter has the following characteristics,

a) Mass (made up of particles)

b) Volume (occupies space)

13. Correct options: (A, B)

The freezing point of water are 0°c (Celsius scale) and 273K (In Kelvin scale)

14. Correct options: (B, C)

Iron rod and wooden box are examples of solid state of matter because they have definite shape and volume as solids state of matter.

15. Correct option: (A)

Plasma is a distinct state of matter because, at very high temperatures, gas particles gain enough energy to lose electrons. This creates a mixture of **positive ions and free electrons**, known as **plasma**. Since plasma has properties different from solids, liquids, and gases, it is recognized as another state of matter (often called the fourth state, though some texts refer to it as the fifth state).Therefore, the reason correctly explains why plasma is considered a separate state of matter.

16. Correct option: (A)

In solids, particles are arranged very closely with strong intermolecular forces. Because they are tightly packed, the space between molecules is extremely small (negligible). Thus, the reason directly explains the assertion.

17. Correct option: (C)

Melting is indeed the change of a **solid into a liquid** by absorbing heat.

The **melting point** is the temperature at which a **solid changes into a liquid**, not a liquid into a gas.

The temperature at which a **liquid changes into a gas** is called the **boiling point**.

18. Correct option: (C)

Matter is correctly defined as anything that has **mass** and occupies **space**.

According to the particle theory of matter, particles are **always in motion**.

In solids, particles vibrate about fixed positions.

In liquids and gases, particles move more freely.

Therefore, the statement that particles are not in motion is incorrect.

19. Correct option: (A)

Condensation is the change of state from **gas to liquid**.

During condensation, gas particles **lose heat energy** to the surroundings, move more slowly, and come closer together to form a liquid.

The reason correctly describes what happens during condensation: when gases are cooled, they lose heat and can change into liquids.

20. Correct Matching is: a-P, b-Q,c-S, d-R

a) On increasing the temperature substance convert from liquid state to gas is known as evaporation.

b) On cooling substance present in gas form convert into liquid state is known as condensation.

c) Conversion of solid substance into gas state (without converting into liquid state) is known as sublimation.

d) On heating solid substance changes to liquid is known as fusion.

21. Correct Matching is: a-P,b-Q,c-S,d-R

a) Matter is anything made up of molecules or atoms.

b) Amorphous is solids with random arrangement of particles.

c) Charged particles present in plasma state of matter.

d) Solid with a particular geometrical arrangement of particles is known as crystalline.

22. Correct option: (B)

Liquids do not have a fixed shape, but they have a fixed volume. Therefore, water takes the shape of the container (glass) in which it is kept.

23. Correct option: (C)

The smell of coffee spreads through the air because gas particles move freely and mix with other gases. This process is called **diffusion**. It shows that gases spread easily in all directions.

24. Correct option: (C)

Chalk is a **solid**. In solids, particles are very closely packed and can only vibrate in their positions. Because of this, solids have a fixed shape and do not change shape easily.

25. Correct option: (B)

Sublimation is the process in which a **solid changes directly into a gas** without passing through the liquid state. Examples include naphthalene balls, camphor, and dry ice.

26. Correct option: (c)

Naphthalene balls undergo **sublimation**. They do not melt first; instead, they change directly from a solid to a gas and gradually disappear.

27. Correct option: (B)

In sublimation:

The solid changes directly into its gaseous form, called the **sublime**.

When the gas cools and changes back into a solid, the process is called **deposition**

28. Correct Answer is: 7

we know that, 0°c=273K

Given280K=273K+7

=0°c+7=7°c

Therefore 280K=7°c

29. Correct Answer is: 3

Matter mainly available in three states as solids, liquids and gaseous state.

30. Correct Answer is: 2

Mixture is divided into two types; they are homogeneous mixture and heterogeneous mixture.

Elements, compounds and mixtures

Chemical classification of matter

LEVEL-I

1. Correct option: (A)

Pure substance is homogeneous in nature.

2. Correct option: (A)

Elements exist in all three states. Fluorine exists in gaseous state.

3. Correct option: (C)

Mercury (Hg) exists in liquid state.

4. Correct option: (D)

Carbon is the element that present in both diamond and graphite.

5. Correct option: (B)

Mercury is the liquid metal at room temperature.

6. Correct option: (A)

Selenium is the metalloid that shows both the properties of metals and non-metals

7. Correct option: (C)

Metalloids have the properties of both metal and non-metal.

8. Correct option: (C)

Radium (Ra) is one of the radioactive elements that emit harmful radiations.

9. Correct option: (D)

Sodium (Na) is the metal.

10. Correct option: (B)

Sulphur is nonmetal and remaining all shows the properties of metals.

11. Correct option: (A)

Metals are generally malleable and ductile. These are properties shown by metals.

12. Correct option: (B)

Metals form cations by losing the electrons. This property is known as electro positivity nature of metals.

13. Correct option: (C)

The property of metal, converting into thin sheet by beating on the metal is known as malleability.

14. Correct option: (C)

All non-metals are bad conductors of heat and electricity except graphite and gas carbon.

15. Correct option: (B)

Metals have electro positive valency, because of that they easily loose electrons and forms cations.

16. Correct option: (B)

Non-metals have electro negative valency, because of that they accept electrons and forms anions.

17. Correct option: (C)

Metals have the property of ductility. Al belongs to metals group and it shows ductile property where as remaining all belongs to nonmetals.

18. Correct option: (C)

In general, all metals react with acid to form salt and hydrogen gas. The reactions are also known as displacement reactions.

19. Correct option: (A)

These properties are shown by the metals. Some of the properties are sonority (produce sound), lustre (shining), conductivity, ductility etc.

20. Correct option: (B)

The property of a metal to raise sound is known as sonority. This is one of the properties that is shown by metals.

21. Correct option: (B)

The materials that allow electricity to pass through them are called conductors. This is also one of the properties that shown by metals.

22. Correct option: (C)

The materials that do not allow electricity to pass through them are called insulators. This is also one of the properties that shown by non-metals.

23. Correct option: (B)

When nonmetal oxides dissolve in water, they can react with the water molecules to form acids. The oxides combine with water to produce acidic solutions by releasing hydrogen ions. So, the oxides of nonmetals are acidic.

24. Correct option: (C)

Metallic oxides in general tend to be basic in nature. This is because metals have a tendency to lose electrons and form cations, resulting in the formation of ionic compounds when combined with oxygen.

25. Correct option: (C)
Because of electro positivity of metals, they liberate hydrogen when reacts with acids.

26. Correct option: (C)
In the properties of metals, we already know that metals are good conductors of heat and electricity.

27. Correct option: (C)
Radon is the naturally occurring radioactive element.

28. Correct option: (D)
Radium is an example of radioactive element.

29. Correct option: (A)
Graphite is a nonmetal that conducts electricity because of free electrons, whereas all nonmetals are bad conductors of electricity.

30. Correct option: (C)
Metals forms basic oxides; it does not form acidic oxides. Remaining all are properties of metals.

LEVEL-II

1. Correct option: (C)
Gold and platinum are the Nobel metals which do not rust because they do not react with oxygen.

2. Correct option: (A)
Based on the reactivity order metals can displace each other. Less reactivity metal is replaced by more reactivity metal.

3. Correct option: (D)
Nonmetals do not have lustrous property, but Iodine is exception which has the lustrous property.

4. Correct option: (A)
Caesium is the metal that has low melting and boiling because of big atomic size and weak intermolecular forces of attraction.

5. Correct option: (D)
Because of carbon bonding and its three-dimensional structure diamond is the hardest substance even it is a nonmetal(C).

6. Correct option: (D)
In general, all non-metals are bad conductors of electricity. In the given elements Phosphorus is the only elements belongs to non-metals and the remaining all belongs to metals group.

7. Correct option: (D)
In given substances Duraluminum, Magnalium, bell metal are formed by the combination of two or more elements physically (Alloys) comes under impure substances and the remaining element Magnesium comes under pure substances which contains only one type of atoms.

8. Correct option: (B)

The property of metals by which they can be beaten into thin sheets is known as malleability. So here Aluminium foil can be made from aluminium by using its malleability property.

9. Correct option: (B)

Sulphur is the polyatomic exist as S_8 is more stable from and the remaining given elements exists as diatomic.

10. Correct option: (D)

In general metals are good conductors of electricity. In the given elements only, Aluminium belongs to metals and the remaining all elements belongs to non-metals. So Aluminium is the good conductor of electricity.

11. Correct option: (A)

Diamond is the hardest substance because of its carbon bonding.

12. Correct options: (A, C)

Because of high resistivity, Tungsten and Bismuth exhibits poor conductivity of electricity compared to other metals that exhibits more electrical conductivity.

13. Correct options: (A, B, C, D)

All the given statements are correct,

14. Correct options: (A, B)

A. pure substance contains only one type of elements, it is not heterogeneous in nature.

B. The composition of a pure substance will remain same, cannot be altered by any phenomena.

15. Correct options: (A, C, D)

Only kerosene contains one type of compound where as the remaining three contains more than one compounds formed by mixing more than one compound physically.

16. Correct options: (A, B, D)

HCl exist as liquid and the remaining three are gaseous compounds.

17. Correct options: (B, D)

Brass and steel are alloys, which are the examples of homogeneous mixtures.

18. Correct option: (C)

A pure substance contains **only one kind of particle** (element or compound) and has a uniform composition throughout, so it is homogeneous. The reason is false because different kinds of particles mixed together form a **mixture**, not a pure substance.

19. Correct option: (C)

Mercury is a metal that remains liquid at room temperature. Most metals are solids at room temperature, not gases. Therefore, the reason is incorrect.

20. Correct option: (C)

Metalloids such as silicon and germanium show characteristics of both metals and non-metals. Radioactive elements actually **do emit harmful radiations**, so the reason is false and unrelated to the assertion.

21. Correct option: (A)

A heterogeneous mixture has a non-uniform composition. Different portions of the mixture contain different amounts of the components. Thus, the reason correctly explains the assertion.

22. Correct option: (B)

Both statements are true. However, the reason describes **how compounds can be separated**, not **why compounds are formed in fixed proportions by weight**. Therefore, it does not correctly explain the assertion.

23. Correct Matching is: a-T,b-S,c-Q,d-P

a) In given elements Iron is Metal

b) Carbon is non metal

c) Arsenic is metalloid

d) Radioactive element is Polonium

24. Correct Matching is: a-P,S,T;b-Q,c-R,d-s

a) Heterogeneous mitures, in which two or more components are not mixed uniformly. soil, oil+water, sand and salt comes under these types of mixtures.

b) Copper+zinc forms an alloy of Brass.

c) Homogenous mixture, in which the components are uniformly mixed, solution is an example of this mixture.

d)Immiscible liquid, in which the components formed two separate layers after certain time of mixing when they are not disturbed. Oil+ water comes under this liquid.

25. Correct option: (B)

Nitrogen is a non-metal found in the atmosphere.

Iron, Aluminium, and Silver are all metals.

26. Correct option: (A)

Electropositive elements tend to **lose electrons** and form positive ions. Metals are generally electropositive.

Sodium is a highly electropositive metal.

Nitrogen, Chlorine, and Fluorine are non-metals and are electronegative.

27. Correct option: (D)

Metalloids show characteristics of both metals and non-metals.

Examples: Silicon, Germanium, Arsenic.

28. Correct option: (B)

Tap water contains dissolved minerals, salts, and gases, making it a **mixture** rather than a pure substance.

29. Correct option: (B)

Milk contains water, fats, proteins, vitamins, and minerals mixed together, so it is a **mixture**.

Oxygen is an element.

Gold is an element.

Iron is an element.

30. Correct option: (C)

Medicines are prepared by combining different chemical substances in specific proportions to achieve the desired therapeutic effect

31. Correct Answer is: 6

2. Materials – Sorting and Separation

Solutions

Properties of Materials

LEVEL - I

1. Correct option: (A)

A thing that can be seen and touched is called an object.

2. Correct option: (C)

The matter of which an object is made is called material.

3. Correct option: (B)

The sorting of objects into groups, with each group having its own characteristic properties is called classification of objects.

4. Correct option: (C)

Baby is the odd one. Baby is living thing where as the remaining all are non-living things.

5. Correct option: (C)

In given examples boat is non-living thing where as the remaining all are living things **6.Correct option: (B)**

The two types of materials are natural and man made materials.

7. Correct option: (B)

Transparent materials allow **almost all light to pass through**, so objects behind them can be seen clearly.

Air – Transparent

8. Correct option: (D)

In the given example Coal is natural material and the remaining all are manmade materials

9. Correct option: (A)

Nylon is prepared by humans where as the remaining cotton, silk and cotton are obtained from the nature.

10. Correct option: (A)

Ground glass allows some light to pass through but does not allow objects to be seen clearly. Therefore, it is translucent, not opaque. Brick, book, and wood are opaque materials.

11. Correct option: (A)

Ice has a lower density than liquid water, so it floats on water. Aluminium, chalk powder, and iodine are denser than water and sink.

12. Correct option: (A)

Wood is generally less dense than water, so it floats. Soil, chalk powder, and glass are denser than water and sink.

13. Correct option: (A)

Chalk powder is insoluble in water and forms a suspension. Alum, washing powder, and blue vitriol (copper sulphate) dissolve in water.

14. Correct option: (A)

Kerosene does not mix with water and forms a separate layer because it is immiscible. Vinegar, glycerine, and lemon juice mix completely with water.

15. **Correct option: (A)**
Lead is toxic and harmful to health, so it is not used for making cooking utensils. Copper, stainless steel, and aluminium are commonly used.

16. **Correct option: (A)**
Iron reacts with oxygen and moisture to form rust. Rusting causes the surface to lose its shine.

17. **Correct option: (A)**
Hardness is the property that indicates a material's resistance to scratching. Hard materials are difficult to scratch, while soft materials are scratched easily.

18. **Correct option: (A)**
Sugar dissolves completely in water to form a clear, homogeneous solution. Sand, chalk powder, and sawdust do not dissolve.

19. **Correct option: (A)**
Opaque materials do not allow light to pass through, so objects behind them cannot be seen. Transparent materials allow light to pass completely, while translucent materials allow only some light to pass.

LEVEL-II

1. **Correct option: (D)**
The hardest material is Diamond.

2. **Correct option: (B)**
Hard materials resist scratching, cutting, bending, and compression. Examples include iron, stone, and diamond.

3. **Correct option: (A)**
Soft materials can be easily cut, bent, scratched, or compressed. Examples include rubber, sponge, and cotton.

4. **Correct option: (D)**
Sugar, salt, and copper sulphate dissolve in water, whereas sand is insoluble in water. Therefore, sand is the odd one out.

5. **Correct option: (B)**
Algae contain chlorophyll and prepare their own food by photosynthesis using carbon dioxide, water, and sunlight. Animals cannot make their own food.

6. **Correct option: (C)**
A feather is a non-living object because it does not grow, reproduce, or perform life processes independently. Fish, fowl, and fox are living organisms.

7. **Correct option: (B)**
Leather is a processed material made from animal skin. Minerals, iron, and gold occur naturally in the Earth.

8. **Correct option: (B)**
A crane is a bird (animal). Cucumber and cactus are plants, and Croton is also a plant.

9. **Correct option: (A)**
Clouds allow only some light to pass through and do not allow clear vision through them, so they are translucent. Cardboard is opaque, while clear glass and groundnut oil are transparent.

10. **Correct option: (D)**
Diamond is the hardest naturally occurring substance. It is used in cutting and drilling tools because of its exceptional hardness**.**

One or more than one option correct type:

11. Correct option: (A),(B),(C)

Soft materials can be easily compressed, bent, cut, or scratched.
Sponge is soft and can be compressed easily.
Cotton is soft and flexible.
Rubber is generally soft and can be bent or compressed.
Iron is a hard material and cannot be easily scratched or compressed.

12. Correct option: (A),(C),(D)

Insoluble substances do not dissolve in water.
Sand does not dissolve in water.
Sawdust does not dissolve and floats on water.
Paper pieces do not dissolve in water (though they may become soggy).
Salt dissolves completely in water, so it is soluble**.**

13. Correct option: (A),(C),(D)

Transparent materials allow light to pass through them, so objects behind them can be seen clearly.
Air is transparent.
Glass (clear glass) is transparent.
Water (pure and clean) is transparent.
Wooden box is opaque because it does not allow light to pass through.

Assertion and reason type:

14. Correct option: (A)

A freshly cut piece of sodium has a shiny metallic surface.
However, sodium is highly reactive and quickly reacts with oxygen and moisture present in the air to form a dull coating of sodium oxide and sodium hydroxide.
Therefore, the freshly cut surface appears shiny only for a short time.

15. Correct option: (A)

Translucent materials allow **some light** to pass through.
The transmitted light is **scattered**, so objects cannot be seen clearly through them.
Hence, the reason correctly explains why translucent materials provide only partial visibility.

Matching:

16. a) Glass →Q) Transparent material
Clear glass allows light to pass through completely.
b) Cotton→P) Soft material
Cotton is soft and can be compressed easily.
c) Iron→R) Hard material
Iron is hard and cannot be scratched or bent easily.
d) Oiled paper→S) Translucent material
Oiled paper allows some light to pass through but objects are not seen clearly.

Comprehension:

17. Correct option: (C)

Sugar dissolves completely in water to form a clear solution. Therefore, it is a soluble substance**.**

18. Correct option: (C)

Sand does not dissolve in water and is denser than water. Therefore, it sinks and settles at the bottom of the container.

19. Correct option: (C)
Oil does not mix with water because it is insoluble. Since its density is lower than that of water, it floats on the surface.

Materials – Separation

LEVEL-I

1. Correct option: (C)
The process of thrashing involves that the stacks were thrashed against a slab of rock.

2. Correct option: (A)
The process that is used by framers for separating husk from grains is known as winnowing.

3. Correct option: (D)
The grains from the stalks were separated by thrashing process.

4. Correct option: (B)
The process of separating visible unwanted particles by hand is called handpicking.

5. Correct option: (D)
The process of separating the particles of different size by using a sieve is called sieving.

6. Correct option: (D)
We can separate insoluble solid particles in liquid by using filtration process .in the given examples sand dissolved in water can be separated by using filtration, because sand is insoluble solid particles in water (liquid).

7. Correct option: (B)
In soluble solid material which settles down in the sedimentation process is known as sediment.

8. Correct option: (D)
The process of settling down heavier and insoluble components from a mixture in a liquid is called sedimentation.

9. Correct option: (B)
The process of separation of an insoluble solid constituent of a mixture from its liquid component is called filtration.

10. Correct option: (C)
The process of separation of impurities settled at the bottom when muddy water was kept overnight in a bucket and the pure water was then poured off from the top is known as decantation.

11. Correct option: (C)
In common salt solution, the constituents were separated by using evaporation (dissolved solid present in liquid).

12. Correct option: (B)
The process of changing a liquid into gaseous state on heating is called evaporation.

13. Correct option: (A)
The process of separation of liquid from a soluble salt solution by the evaporation of liquid and recondensation of its vapors in another level is called distillation.

14. Correct option: (B)
The condenser that is used for recondensation the gaseous molecule in distillation process is known as Liebig's condenser.

15. Correct option: (D)
In sublimation process, the solid state formed from the gaseous state on cooling is called sublime.

16. Correct option: (A)
Iodine on heating directly converts from solid state to gaseous state in sublimation process. So, this process is used for separation of sand and Iodine mixture.

17. Correct option: (C)
The process of subliming a solid from a mixture by heating is called sublimation.

18. Correct option: (A)
The process of separation of suspended particles in a liquid by whirling at very high speed is called centrifugation.

19. Correct option: (C)
Blood and other biological samples are separated by using centrifugation. By using this process, the suspended particles will settle at the bottom because of the high-speed whirling.

20. Correct option: (B)
The method used to separate cream from curd is centrifugation. In these the suspended cream particles will be separated when the mixture is whirled at high speed.

LEVEL-II

1. Correct option: (C)
The spoiled and bad vegetables are removed from the fresh vegetables by hand picking method.

2. Correct option: (B)
Jewelers used sieving method to separate various sizes of pearls.

3. Correct option: (D)
Evaporation process is used to make large amount of common salt from seawater. In this process liquid gets evaporated to separate soluble salt on heating and salt will remains at the bottom of the flask.

4. Correct option: (B)
In the process of evaporation, a liquid change into gaseous state on gentle heating, such that the liquid does not boil.

5. Correct option: (D)
In gulf countries the salt water (saline water) is converted to fresh water by an artificial process. The process is called desalination or distillation.

6. Correct option: (B)
The principle behind the filtration is the solid component present in the liquid, which is insoluble in liquid.

7. Correct option: (D)
Since silver chloride is insoluble in water, the mixture can be separated by filtration. That is by passing through filter paper.

8. Correct option: (A)
Compared to sedimentation and decantation, filtration is more beneficial, because in filtration process the constituents of the mixture of a solid and liquid can be separated completely.

9. Correct option: (B)

In given examples, cornflakes in milk can be separated by using filtration. Because for filtration process the mixture should contain insoluble solid constituents in liquid.

10. Correct option: (C)

A mixture of tea leaves and milk come under the examples of insoluble solid constituents present in liquid and which is separated by using filtration process.

11. Correct option: (B)

Sand and iron fillings are separated by using a magnet, because iron filling are easily attracted towards magnet and can be separated easily.

12. Correct option: (B)

In the given mixture salt is dissolved solid, that present in liquid. These types of mixtures can be separated by using evaporation, where water is evaporated.

13. Correct options: (A, B)

We can separate certain solid-solid mixtures by using solvents in which one component is soluble and one component is not soluble in a specific solvent.

14. Correct options: (A, B, D)

In the following statements, solid –solid mixtures can be separated by using magnetic separation, gravity method and sublimation method.

15. Correct options: (A, B, C, D)

All statements are correct.

Assertion and reason type:

16. Correct option: (A)

Assertion: True. Sieving cannot separate flour dissolved or mixed in water because one component is a liquid.

Reason: True. Sieving is used to separate solid particles of different sizes, such as flour and bran.

Since flour and water form a solid–liquid mixture, filtration is suitable, not sieving. Thus, the reason correctly explains the assertion.

17. Correct option: (A)

Assertion: True. Distillation separates pure water from dissolved salts.

Reason: True. During distillation, water evaporates first and is then condensed into pure liquid, while dissolved salts remain in the flask.

Therefore, the reason correctly explains the assertion.

18. Correct Matching is: a-S; b-Q; c-P; d-R

19. Correct Matching is: a-S; b-Q; c-P; d-R

COMPREHENSION TYPE

20. Correct option: (B)

Winnowing separates lighter husk from heavier grains using wind. The method works because of the difference in weight (or density) between the two components.

21. Correct option: (C)

A filter has tiny pores that allow the liquid to pass through while retaining insoluble solid particles.

22. Correct option: (C)

Heating causes the solvent (usually water) to evaporate as vapour, leaving the dissolved

solid behind. This method is called evaporation.

23. Correct Answer is: 1

24. Correct Answer is: 3

Previous year

1. Correct option: (B)

Transparent Materials

These materials allow light to pass through them completely, which means you can see through them clearly.

Magnifying glass (v): Made of clear glass or plastic designed to let light pass cleanly so you can view objects closely.

Fresh air (vi): Completely clear gas that lets light travel unhindered.

2. Translucent Materials

These materials allow light to pass through only partially. Objects on the other side appear blurred or faint.

Butter paper (iii): A semi-transparent paper that scatters light, making it translucent.

3. Opaque Materials

These materials do not allow any light to pass through them. You cannot see through them at all.

Rubber ball (i): Solid material that blocks light completely.

Newspaper (ii): Thick paper printed with ink that absorbs/reflects light rather than letting it pass.

Leather bag (iv): A thick, dark, and solid organic material that completely blocks light.

4. Correct option: (A)

(i) Polythene→ (c) Easily moulded: Polythene is a thermoplastic material, meaning it softens upon heating and can be easily shaped or moulded into various forms (like bags or containers).

(ii) Gold coin →(e) Sinks in water: Gold is a highly dense metal. Because its density is much greater than that of water, a gold coin will sink immediately.

(iii) Sugar powder →(d) Soluble in water: When you mix powdered sugar into water, the sugar molecules disperse uniformly between the water molecules, completely dissolving.

(iv) Feather →(b) Floats on water: Feathers are very light and trap air, giving them a very low effective density, which allows them to float on the surface of water.

(v) Frosted glass →(a) Translucent: Frosted glass has a rough surface that scatters light passing through it. Because it only allows light to pass through partially (making objects behind it look blurry), it is classified as translucent.

5. Correct option: (D)

Object P must be: Opaque,Sinks in water and Lustrous

Lustrous means it is shiny. Metals like copper and gold are lustrous, while a book or oil is not. This eliminates options A and B.

Looking at the remaining choices (Copper rod vs. Gold ring), both fit all three criteria perfectly.

Object Q must be: Opaque Does NOT sink in water (x i.e., it floats), and is NOT lustrous (x).

Let's check the options left from C and D for Object Q:

Option C (Coin): A metal coin is lustrous and sinks in water, which violates both conditions.

Option D (Wooden block): A wooden block is opaque, it floats on water (does not sink), and it is non-lustrous. This fits perfectly.

Object R must be: Opaque Sinks in water and is NOT lustrous (x).

Let's check the Object R option for row D:
Brick: A brick is opaque, it is dense so it sinks in water, and it has a dull, non-shiny surface (not lustrous). This perfectly satisfies all conditions.

6. **Correct option: (C)**
Statement I: *"Components will disappear completely in test tubes L and M."* * **Incorrect.** The component in test tube L is sand, which does not dissolve or disappear; it stays visible at the bottom.
Statement II: *"Components in test tubes L, M and N are completely soluble in water."*
Incorrect. As established, sand (L) and chalk powder (N) are insoluble.
Statement III: *"Components in test tubes K, M and O are insoluble in water."*
Incorrect. Sugar (K), lemon juice (M), and honey (O) are all actually **soluble** in water.
Statement IV: *"Components in test tubes K and O will disappear completely."*
Correct. Since sugar (K) and honey (O) are soluble, they dissolve and disappear completely into the water matrix.

7. Correct option: (C)
C is correct because hardness is a relative property; materials that are soft (like chalk or wax) can be easily scratched, compressed, or cut compared to hard materials (like iron or diamond).

8. Correct option: (C)
Objects float if they are less dense than water and sink if they are denser than water. A metallic key, coin, and iron nail are all made of dense metals and will sink. A **dry leaf** is incredibly lightweight and less dense than water, so it stays on the surface (floats).

9. **Correct option: (B)**
Let's look closer at Group II: *Food colour, vinegar, lemon juice, and chalk powder*. While food color, vinegar, and lemon juice dissolve perfectly, **chalk powder is completely insoluble** in water and will settle at the bottom. Therefore, this statement is false.

10. **Correct option: (B)**
Magnifying glass →Translucent: Incorrect. It is completely **transparent** so you can see objects through it clearly.
Muslin cloth →Opaque: Incorrect. Muslin cloth is a thin, woven cotton fabric that lets some light pass through, making it **translucent**.
Dressing mirror → Transparent: Incorrect. A mirror has an opaque silver coating on the back that reflects light instead of letting it pass through; it is classified as **opaque**.

11. Correct option: (C)
Section 1 requires a material that is "Immiscible with water":
Glycerine and *Honey* are highly miscible (they mix completely with water).
Vaseline (petroleum jelly) is an oil-based product that does not mix with water at all; it is immiscible. This immediately points us toward **Option C**.
Section 2 requires a material that is "Hard and Lustrous":
Metal spoon: Metals are physically hard and have a shiny, reflective appearance (*lustrous*). This perfectly matches Option C. (Chalk, leather, and sponges are neither hard nor lustrous).
Section 3 requires a material that is "Opaque":

Cardboard: It completely blocks light from passing through, making it fully opaque.

Section 4 requires a material that is a "Liquid, heavier than water":

Mercury: It is a liquid metal at room temperature and is extremely dense ($13.6 g/Cm^3$), meaning it is much heavier than water ($1 g/Cm^3$) and sinks immediately.

12. **Correct option: (B)**

Slide Q (Painted with light blue colour): Coating a transparent glass slide with a thin layer of light-coloured paint transforms it into a translucent material. It allows light to pass through only partially, which means students will see the (X) mark partially or blurred.

3. Language of Chemistry

LEVEL - I

1. Correct option: (B)

J.J.Berzelius is the scientist who suggested a method of representing elements using the English letters (capital as well as small)

2. Correct option: (D)

The symbol C stands for Carbon.

3. Correct option: (A)

One gram atom of He is the mass of one mole of an element equal in grams to the atomic weight. That is molecular mass of substance. That is 4g.

4. Correct option: (B)

One gram atom of oxygen is molar mass of oxygen, i.e., 16g.

5. Correct option: (C)

A) The symbol 'O' represents short hand sign of element oxygen.
B) The symbol represents quantitatively one atom of oxygen
C) one gram atom of oxygen atom is 16 g
D) The atomic number of Oxygen is 8 not 6

6. Correct option: (C)

Hydrargyrum (Hg) is the Latin name of Mercury. The meaning of the name is liquid silver. It is a liquid metal.

7. Correct option: (C)

As per the Berzelius representation, symbol for Silicon is 'Si'.

8. Correct option: (A)

The Latin names for the element 'Copper' and 'silver' are respectively Cuprum, Argentum. So, the symbols for them are Cu and Ag.

9. Correct option: (A)

Latin name of Sodium is Natrium and the symbol is Na.

10. Correct option: (B)

Latin name of Potassium is Kalium and the symbol is 'K'.

11. Correct option: (B)

V stands for Vanadium. which is derived from the first letter of the element name.

12. Correct option: (C)

Cu stands for Cuprum. The symbol is derived from their Latin name.

13. Correct option: (A)

The symbol for lead is derived from the Latin Name Plumbum. Its symbol is 'Pb.

14. Correct option: (C)

The symbol fermium is 'Fm'. It is derived from the Scientist Enrico Fermi.

15. Correct option: (B)

The name for the given symbols are
i) 'Cu' stands for Cuprum
ii) 'Au' stands for Aurum
iii)'Hg' stands for Hydrargyrum
iv) 'Fe' stands for Ferrum. All these symbols were derived from their Latin names.

16. Correct option: (B)

'Ti' is the symbol for Titanium derived from is English name.

17. Correct option: (B)

'Sc 'stands for the element Scandium.

18. Correct option: (A)
Hydrogen is the lightest elements.

19. Correct option: (B)
Hydrargyrum (Hg) is the Latin name of Mercury. The meaning of the name is liquid silver. It is a liquid metal.

20. Correct option: (B)
The symbol of 'Iodine' is I

21. Correct option: (B)
The symbol for chlorine is 'Cl'.

22. Correct option: (C)
The symbol of Aluminum is 'Al'

23. Correct option: (C)
A symbol represents the atom of element.

24. Correct option: (C)
The symbol for Potassium is 'K' derived from the Latin word Kalium.

25. Correct option: (A)
Mendelevium is the element that named after the name of the scientist Mendeleev (some elements are named after the names of the scientists.)

LEVEL -II

1. Correct option: (B)
Latin name of Hg is Hydrargyrum, in Greek Hydro means water and argyros means silver. Hg is a heavy silvery d-block element. The common name of the element is Mercury.

2. Correct option: (D)
Given all statements are correct.

3. Correct option: (D)
The symbol 'Pd' stands for the element Pelladium.

4. Correct option: (A)
The name of the element having atomic number 16 is Sulphur and the symbol is 'S'.

5. Correct option: (B)
For some of the elements the symbols are taken from their Latin names.

6. Correct option: (C)
The element Curium named after the scientist Madam Curie.

7. Correct option: (A)
Uranus is the planet after which the element is named as 'Uranium'.

8. Correct option: (B)
'Po" stands for the element 'Polonium'.

9. Correct option: (B)
The symbol for the element Uranium is 'U'.

10. Correct option: (C)
J.J.Bezelius was the scientist who developed a method of representing elements using the English letters (capital as well as small)

11. Correct options: (A, B, C, D)
All the given element are named from their Latin mames.
A)Lead-Plumbum (Pb)
B) Tin- Stannum (Sn)
C) Antomony –Stibium (Sb)

d) Copper-Cuprum (Cu)

12. Correct options: (A, B, C, D)

All the given elements are named after the name of the Scientists.

	Element	Scientist name
A)	Mendelevium	- Mendeleev
B)	Fermium	- Enrico Fermi
C)	Nobelium	- Alfred Nobel
D)	Curium	- Madam Curie

13. Correct options: (A, B, C, D)

All the given elements are named after the Countries and laboratories of the countries.

A) Berkelium- city of Berkley

B) Californium-University of California

C) Polonium - Poland

D) Ruthenium – Russia

14. Correct options: (A, C)

The symbol 'Au' indicates the Latin name of Gold which is Aurum.

15. **Answer: (B)** Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

16. **Answer: (C)** Assertion is true, but Reason is false.

17. Correct Matching is: a-R,b-P,c-S,d-Q

The symbols for the given elements are,

a) Platinum – Pt

b) Palladium – Pd

c) Mercury – Hg

d) Barium – Ba

18. (D) Ca, Be, Co, Sb

19. (A) Br, Ag, Cu, Mn

20. Answer: (C) Sn

- **Tin → Sn (from Latin *Stannum*)**

21. Correct Answer is: 3

Three elements are names after the names of the elements.

22. Correct Answer is: 7

Atomic number of N is 7.

Valency

LEVEL-I

1. Correct option: (A)

The combining capacity of an element is known as valency.

2. Correct option: (D)

Helium, Neon, Argon belongs inert gases, in which outré most shell is completely filled. When outer most shell is completely filled, the valency will become zero. Because there are no unpaired electrons.

3. Correct option: (A)

The electronic configuration of Na is 2, 8, 1. The no. of valency electrons present in a metal atom is its valency. So, the valency of Na is 1.

4. Correct option: (B)

For nonmetals the valency = 8- no. of valency electrons.

For oxygen electronic configuration is 2,6

Valency electrons=6
Valency=8-6=2

5. Correct option: (A)
In ammonium ion, Nitrogen combines with four hydrogen atoms. So, the valency is equal to no. of hydrogens combines directly. So, the valency of ammonium ion is 4

6. Correct option: (B)
The valency for the compounds combined with chlorine is equal to the no. of chlorine atoms combine directly or indirectly to the compound. In MgCl2 the no. of chlorine atoms combined to form the compound is 2. So, the valency of MgCl2 is 2.

7. Correct option: (C)
Iron (Fe)exhibits variable valencies 2 and 3.for these the compounds in which the elements shows the higher valency is indicate by the suffix 'ic'. While the compound in which the element shows lower valency is indicated by 'ous' compounds. So, the valency of 'Fe' in ferric chloride is 3

8. Correct option: (A)
In the compound MgO,1 atom of magnesium combines with 1 atom of oxygen, therefore valency of magnesium is 2.

9. Correct option: (A)
From the chart of variable valencies, we know that copper is one of the elements that shows variable valencies, those are 1and 2. The lower valency of copper is 1.

10. Correct option: (B)
From the chart of variable valencies, we know that silver is one of the elements that shows variable valencies, those are 1and 2. The lower valency of silver is 1.

11. Correct option: (A)
From the chart of variable valencies, we know that mercury is one of the elements that shows variable valencies, those are 1and 2. The higher valency of mercury is 2.

12. Correct option: (D)
In 'ic' compounds the element shows higher valency. From the chart of variable valencies the valencies that are shown by copper are 1 and 2. In cupric chloride the valency of copper is 2.

13. Correct option: (B)
In 'ous' compounds the element shows lower valency. From the chart of variable valencies 'Fe' shows variables valencies 2 and 3.in ferrous compound the valency of 'Fe' is + 2.

14. Correct option: (C)
In 'ic' compounds the element shows higher valency. From the chart of variable valencies the valencies that are shown by copper are 1 and 2.the charge on cupric ion is +2.

15. Correct option: (D)
The variable valencies that are shown by Tin are 2and 4.stannum shows higher valency 4 by combining with four chlorine atoms to form SnCl4.

16. Correct option: (D)
In the given elements Stannum is the element which shows variable valencies .it shows 2 and 4 (given in the table of variable valencies) where as the remaining elements shows only one valency,

17. Correct option: (A)
The element Aurum shows variable valencies 1 and 3.in 'ous' compound the elements shows lower valency. So the valency in aurous ion is +1.i.e.Au^{+1}

18. Correct option: (D)

The element Aurum shows variable valencies 1 and 3.

LEVEL-II

1. Correct option: (B)

The valency of a nonmetal is =8-number of valency electrons.

2. Correct option: (B)

Some of the elements show more than one valency. The compound in which the element shows higher valency is indicated by the suffix 'ic 'compounds.

3. Correct option: (C)

Some of the elements show more than one valency. The compound in which the element shows lower valency is indicated by the suffix 'ous 'compounds.

4. Correct options: (A, B, C, D)

The elements that are given belongs to inert gas family. In inert gases the outer most shell of the elements is completely filled. So, it has zero vancy.

5. Correct options: (A, C)

For metals valency = no. of outermost electrons

Al, Mn are metals. So, valency for Al is 3 and Mn is 2

For nonmetals valency=8-no. of valency electrons

Nitrogen and carbon are nonmetals

Valency of nitrogen=8-5=3

Valency of carbon=8-4=4

So, the elements with valency 3 are Al, N

6. Correct options: (A, D)

For metals valency=no. of outermost electrons

In given elements Na, Mg belongs to metals

Valency of Na =1 and for Mg=2

For nonmetals valency=8-no. of valency electrons

Valency of Sulphur=8-6=2

Valency of Cl=8-7=1

So, the elements with valency 1 are Na, Cl

7. **(B)** Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

8. Answer: (C) Assertion is true, but Reason is false.

9. Correct Matching is: a-S,b-Q,c-P,d-R

a) Silicon valency is 4

b) Calcium valency is 2

c) Potassium valency is 1

d) Aluminum valency is 3

We know valency of metal=no. of valency electrons

Valency of nonmetal=8-valencece electrons of element.

10. Correct option: (C)

The 'Fe' ion shows variable valencies 2 and 3.the given compound is ferric chloride. In 'ic' compounds it shows higher valency. So the valency of Fe is 3.

11. Correct option: (C)

The valency of N in ammonia is equal to the no. of hydrogens directly combines with nitrogen. So, the valency is 3.

12. Correct option: (B)

In the compound having oxygen the valency is equal to the double the number of oxygen atoms which combine directly or indirectly combines with one atom of an element to form compound. In SO_3, the valency is 2*3=6

13. Correct option: (A)

The valency of Ag in AgCl=no of chlorine atoms combined=1

The valency of $AgCl_2$=No. of Chlorine atoms combined=2

So, the valencies of Ag are 1and 2.

14. Correct option: (A)

The variable valencies of Tin are 2 and 4

In SnCl2 it shows valency 2 and in SnCl4 Sn shows the valency 4.

15. Answer: 3

Aluminium forms **Al^{3+}** ions in aluminium chloride ($AlCl_3$).

Each chlorine has a valency of **1**, so one aluminium combines with three chlorine atoms.

16. Answer: 2

"Ferrous" indicates iron is in the **+2 oxidation state (Fe^{2+})**.

Therefore, the valency of iron in $FeSO_4$ is **2**.

Ions

LEVEL-I

1. Correct option: (C)

An ion is formed by losing or gaining electrons from an atom. So, the no. of electrons will differ an atom from its ion.

2. Correct option: (C)

In the given examples, SO_4^{-2} ion consists of two atoms of different elements to form a single unit. So, it will come under compound ion.

3. Correct option: (D)

In the given examples, only S^{2-} consists of one type element. So, it's a simple ion.

4. Correct option: (B)

If an element loses two electrons, then that element is known as bivalent positive ion. So Mg^{2+} is a bivalent positive ion.

5. Correct option: (B)

If 'Fe" loses two electrons it will become bivalent positive ion and it it denoted by the symbol 'Fe^{2+}'.

6. Correct option: (A)

'Fe' shows variable valencies 2 and 3.the ion with lower valency is known as ferrous ion and it is denoted by 'Fe^{+2}'.

7. Correct option: (A)

Cu shows valencies 1 and 2. The ion with higher valency is known as cupric ion. So, charge on it is 2.

8. Correct option: (A)

The symbol of Chromium ion is 'Cr^{3+}'.

9. Correct option: (D)

The ions which are formed by the loss of three electrons are known as trivalent positive ions. In the given examples all elements belongs to trivalent positive ions.

10. Correct option: (D)

Bromide ion is formed by the gain of one electrons and it is denoted by Br^{-1}. So, the charge on the ions is -1.

11. Correct option: (C)
The symbol of Carbonate ion which is formed by the gain of two electrons is CO_3^{2-}.

12. Correct option: (B)
Sulphate ion (SO_4^{2-}) it is formed by gaining two electrons. So, it is a bivalent negative ion.

13. Correct option: (C)
Acetate ion is formed by the gain of one electron. It is denoted by the symbol CH_3COO^{-}.

14. Correct option: (A)
The name for SO_4^{2-} is sulphate ion.

15. Correct option: (B)
The formula for dichromate ion is $Cr_2O_7^{2-}$.

16. Correct option: (D)
Hydrogen atom by gaining one electron it becomes Hydride ion and it is denoted by H^{-1}. The charge on the ion is -1.

17. Correct option: (C)
The symbol of Hydroxide ion is OH^{-}.

18. Correct option: (A)
The element Iodine becomes Iodide by the gain of one electron and it is denoted by I^{-1}. The charge on the ion is -1.

19. Correct option: (B)
The element Phosphorous become Phosphide by the gain of three electrons and it is denoted by P^{3-}. The charge on it is -3.

20. Correct option: (A)
In tetravalent ions it has a charge 4 on it. In the given ions C^{4-} is the tetravalent ion having 4 as charge.

LEVEL-II

1. Correct option: (A)
The symbol of Zincate ion is given as ZnO_2^{2-}.

2. Correct option: (C)
The symbol of Nitride ion which is formed by the gain of three electrons is N^{3-}.

3. Correct option: (B)
The symbol of Hydrogen sulphite is HSO^{-}_{3}.

4. Correct option: (D)
Symbol of chromate is CrO_4^{2-}.

5. Correct options: (B, C)
Cations are formed by the loss of electrons. These ions have positive charge on it.

6. Correct options: (A, B, C)
Anions are formed by the gain of electrons. These ions have negative charge on it.

7. Correct options: (A, B)
Anions are ions having negative charge on it. In the given ions CH3 COO^{-}. $S2^{-}$ are examples of anions.

8. Correct options: (A, B, C)
The ions that are formed by the gain or lose of three electrons are known as trivalent ions. In the given examples N^{3-},$P^{3-}$and Al^{3+} are trivalent ions.

9. Correct options: (B, D)
In the given electrons O and S have 6 electrons in outermost shell. It required 2 more electrons to get stability. So, these elements gain two electrons to get stability.

10. Correct options: (A, C)

Monovalent anions are those having one negative charge on it. In the given examples OH^-, HSO^-_4 are monovalent ions.

11. Answer: (C) Assertion is true, but Reason is false.

12. Answer: (D) Assertion is false, but Reason is true.

13. Correct Matching is: a-P,b-S,c-Q,d-R

a) Magnesium ion is formed by the loss of two electrons to get stability, so it is bivalent.

b) Aluminum ion is formed by the loss of three electrons to get stability, so it is trivalent.

c) Potassium loss one electron to get stability, so it is monovalent.

d) Carbon loss or gain four electrons to get stability. So, it is tetravalent.

14. Correct Matching is: a-s,b-P,c-Q,d-R

a) cation is formed by loss of electrons

b) Anion is formed by gaining of electrons.

c) Metals are electropositive

d) Nonmetals are electronegative.

15. Correct option: (A)

Cations are positively charged ions. In the given examples Ag+ is cation.

16. Correct option: (C)

In the given ions, an ion which contains more than one atom of the same element is O_2^2.

17. Correct option: (B)

SO_4^{2-} contains two different types of atoms to form a single unit, which will come under the example of compound ions.

18. Correct Answer is: 1

Potassium loss one electron and become monovalent electro positive ion. The charge for monovalent is one.

19. Correct Answer is: 3

Aluminum loss three electrons and become trivalent. So, the charge on it is 3.

20. Correct Answer is: 2

The positive charge on Beryllium is 2. (the number of valency electrons are 2)

Chemical Formula or Molecular Formula

LEVEL-I

1. Correct option: (C)

A chemical formula indicates the exact number of atoms present in molecule.

2. Correct option: (A)

If u observe the chemical formula, more electronegative atom is written on the right side.

3. Correct option: (B)

The chemical formula for Carbon monoxide is CO (where carbon combines with one oxygen atom)

4. Correct option: (A)

The common name for vinegar is 'acetic acid'.

5. Correct option: (B)

The formula for Glucose is $C_6H_{12}O_6$.it is formed by the combination of 6 carbon atoms,12 hydrogen atoms and 6 oxygen atoms.

6. Correct option: (B)

The symbols of aluminum and sulphate ions with their valencies are

Al ion valency is 3

Sulphate ion valency is 2

Now, 2 get associated with Al and 3 gets associated with SO_4. So the final formula of the compound aluminum sulphate is $Al_2(SO4)_3$.

7. Correct option: (C)

The valence shell of A has 3 electrons.

So, valency of A =3

The valency shell of B has 6 electrons.

So, valency =8-valency electrons=8-6=2

So, the valency of A associated with B and Valency of B associated with A as per crisscross method and the symbol is A_2B_3.

8. Correct option: (A)

Given X valency is 2

Valency of Y is 1

In order to form a compound valency of X associated with Y and valency of Y associated with X. Then the formula of compound is XY_2.

9. Correct option: (B)

The valency of X is 4

The valency of Y is 2

In order to form the compound, the valency of X is associated with y and the valency of Y is associated with X, So the formula of the compound is X_4Y_2.

But the valencies 2 and 4 have a common factor 2. So, on dividing the whole formula by 2, we get simplest formula of XY_2. So, the formula is XY_2.

10. Correct option: (B)

'Sn' shows the variable valencies both 2 and 4. In $SnCl_4$ is showing the valency 4. For the compound having higher valency the name will be Stannic chloride.

11. Correct option: (B)

In simple radicals, it contains one or more atoms of the same element. So, in the given examples O_2^{2-} is the simple radical.

12. Correct option: (A)

Metals are electro positive ions. So, Mg will act as electro positive ion and SO_4^{2-} will be the negative radical of the given compo und.

13. Correct option: (B)

As oxygen as negative charge 2 and the element oxygen bonded with two ion of Metal 'M' atom the charge on the M is +1. So, the formula for its chloride is 'MCl' where Cl has negative charge -1.

14. Correct option: (A)

In the given examples, Zn^{+2}, OH^- belongs to simple and compound radicals respectively. Because Zn^{+2} has only one type of elements is example for simple radical and OH^- has two or more different element, which is an example of compound radical.

15. Correct option: (A)

The valency of Calcium is 2.

The valency of Oxygen is 2, by crisscross method

The formula for calcium oxide is CaO. From the formula we can say that the compound has only one oxygen atom.

16. Correct option: (D)

The valency of Calcium is 2.

The valency of Hydrogen is 1

So, two hydrogen atoms combine with one Ca to form calcium hydride. The formula for calcium hydride is CaH_2.

17. Correct option: (A)

'Cu' shows variable valencies 1 and 2. In 'ic' compounds Cu shows higher valency. So, the valency of Cu in cupric chloride is 2and valency of Chloride is =8-7=1. So, 2 associate with chlorine and 1 associate with Cu. So, the formula of Cupric chloride is $CuCl_2$.

18. Correct option: (C)

The valency of Ca is 2 and the valency of PO_4 is 3. The valency of Ca is associated with PO_4 and the valency of PO_4 is associated with Ca. So, the formula is $Ca_3(PO4)_2$.

19. Correct option: (C)

The name of the compound $NaHSO_4$ is Sodium bisulphate.

20. Correct option: (D)

The species $X^{+3}Y^{2-}$ combined to form X_2Y_3, where the valency 3 of X is associated with Y and the valency 2 of Y is associated with X to form the compound.

LEVEL-II

1. Correct option: (A)

The valency of A =3

The valency of B=8-valency electrons in B=8-5=3

When A and B combine together, they form A_3 B_3. If we take the common factor 3, then the formula of the compound is AB.

2. Correct option: (B)

The valency of Mg is=2

The valency of Nitrogen is=8- valency electrons

=8-5=3

The valency of Mg associated with N and the valency of N is associated with Mg. So, the formula for Magnesium nitride is Mg_3N_2.

3. Correct option: (B)

The valency of Al is 3

The valency of Sulphur is =8-6=2, where 6 is no. of valency electrons

For forming Aluminum sulphide, the valency of Al is associated with S and the valency of S associated with Al and the formula of the compound is Al_2S_3.

4. Correct option: (A)

The method that is used to write the chemical formula of a compound is known as Criss-Cross method.

5. Correct option: (C)

Glucose contains 6 atoms of Carbons, 12 atoms of hydrogen and 6 atoms of Oxygen.

6. Correct options: (A, B, C, D)

All the given names are compound names of $NaHCO_3$.

7. Correct option: (A)

The Species $X^{+2}$and Y^{-2} combined together to form X_2Y_2.it has the common factor 2.the short form or simplest form of the compound is XY.

Ans –(A)

8. Correct options: (A, C)

For writing the formulae, the valencies or charges must be balanced.

In writing the formulae of a compound, the metal comes first following by nonmetal.

9. **Answer: (C)** Assertion is true, but Reason is false.

10. Answer: (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

11. Correct Matching is: a-P,b-R,c-S,d-Q

Formula for the given compounds are

a) Washing soda-$Na_2CO_3.10H_2O$

b) Baking soda-$NaHCO_3$

c) Common salt-NaCl

d) Blue vitriol-$CuSO_4.5H_2O$

12. Correct option: (B)

The simple compound are made up of two different elements are called binary compounds.

13. Correct option: (C)

To write the formula for compounds, the name of the method is Criss-Cross method.

14. Correct option: (B)

The valency of Chlorine is equal to the number of chlorine atoms combined. So, the valency of Calcium is 2.

15. Correct Answer is: 4

The valency of Carbon is 4

16. Correct Answer is: 3

The total number of atoms present in water molecules is three (one hydrogen atom and two oxygen atoms) and the chemical formula if H_2O.

4. Basic Concepts of Chemistry
Solutions

LEVEL – I

1. Correct option: (B) John Dalton

John Dalton had postulated that all matter is made up of extremely small particles called atoms, in his atomic theory.

Later, the theory was proved wrong after the discovery of subatomic particles like proton, electron and neutron.

2. Correct option: (A) Democritus

Democritus was a philosopher, proposed the name atomos which means uncuttable. Atom was arrived from Greek word.

3. Correct option: (A) Atom

Atom – Atom is a particle that is made up of nucleus and subatomic particles.

Element – Element is a substance that is made up of atoms with same number of protons.

Compound – Compound is a substance that is made up of two or more different elements.

Mixture – Mixture is a material that is made up of two or more different substances which are not chemically bonded.

4. Correct option: (D) Maharshi Kanada

Maharshi Kanada was an Indian sage. He gave his atomic theory 2500 years earlier than Dalton theory. He proposed that particles can not be divided. And he named the invisible particle as 'anu' and 'parmanu'.

5. Correct option: (B) Indivisible

In Greek 'a' means not and 'tomos' means cut. The word 'atomos' means uncuttable or indivisible. The word 'atom' is also coming from 'atomos'.

6. Correct option: (D) All of these

Dalton proposed that atoms can't be divided as atoms are the smallest particles. But later it was proven that atoms are made up of small subatomic particles protons, neutrons, and electrons. Protons and neutrons are present in the nucleus and electrons are present in the orbitals.

7. Correct option: (C) Electrically neutral

Atoms are consisting of three particles neutrons proton and electrons. Neutrons are neutral particles whereas protons are positive, and electrons are negative. Atoms has same number of electrons and protons. Hence, total positive charge is cancelled by total negative charge and atom is electrically neutral.

8. Correct option: (B) Atomic theory

Dalton in 1804 proposed atomic theory. He proposed in this theory that all matter is composed of extremely small particles called atoms.

9. Correct option: (C) Compound

According to Dalton atoms of different elements combined together to form a compound.

For example – $Na + Cl_2 \rightarrow NaCl$

Where Na and Cl are two different elements. And NaCl is a compound.

And sugar ($C_{12}H_{22}O_{11}$) is also a carbohydrate made up of carbon, hydrogen and oxygen. But sugar and glucose are not as complex as protein.

10. Correct option: (B) Atom

Atom is the smallest unit of matter and takes part in chemical reaction. When different atoms combine together by sharing, gaining or losing of electrons,they form a new molecule. Hence atoms take part in chemical reaction.
Elements, compounds and molecules also can take part in chemical reactions, but they are not the smallest unit of matter.

11. Correct option: (C) The presence of equal number of protons and electrons

Atoms are consisting of three particles neutrons proton and electrons. Neutrons are neutral particles whereas protons are positive, and electrons are negative. Atoms has same number of electrons and protons. Hence, total positive charge is cancelled by total negative charge and atom is electrically neutral.

12. Correct option: (B) Atom

Atom can take part into chemical reactions, but they can't be created or destroyed. Chemical reactions are involved in separation, combination or re-arrangement of atoms.

13. Correct option: (D) -1 charge '0' a.m.u

	Charge	Mass in a.m.u
Proton -	+1	1.0072 ≈ 1
Neutron -	0	1.0086 ≈ 1
Electron -	-1	0.000548 ≈ 0

14. Correct option: (C) The protons and neutrons present in atomic nucleus

Nucleon – Nucleons are number of neutrons and protons present in the nucleus of an atom.
Mass number – The total number of neutrons and protons present in the nucleus of an atom is called mass number.
Atomic number – The total number of protons present in an nucleus of an atom is called atomic number.

15. Correct option: (B) 0.000548

The mass of an electron in a.m.u
= (Mass of electrons in kg) / (mass of one a.m.u)
Mass of electron in kg = 9.01×10^{-31} kg
Mass of 1 a.m.u = 1.66×10^{-27} kg
Hence mass of electron in a.m.u
= $(9.01 \times 10^{-31}) / (1.66 \times 10^{-27})$
=0.000548 a.m.u

16. Correct option: (A) Z

Atomic number – The total number of protons present in an nucleus of an atom is called atomic number.
It is represented by 'Z'.

17. Correct option: (A) Electrons and protons

Neutrons are neutral particles whereas protons are positive and electrons are negative. Atom has equal number of electrons and protons. Hence, total positive charge is cancelled by total negative charge and the atom is neutral.

18. Correct option: (C) 9, 19

Mass number – The total number of neutrons and protons present in the nucleus of an atom is called mass number.
Atomic number – The total number of protons present in an nucleus of an atom is called atomic number.

Atomic number of Fluorine = 9,
Number of neutron present in the nucleus of F atom = 10,
Mass number of Fluorine = (10+9) = 19

19. Correct option: (C) 10, 20

Mass number – The total number of neutrons and protons present in the nucleus of an atom is called mass number.
Atomic number – The total number of protons present in an nucleus of an atom is called atomic number.
Atomic number of Neon = 10,
Number of neutron present in the nucleus of Me atom = 10,
Mass number of neon = (10+10) = 20

LEVEL – II

1. Correct option: (B) 11

No. of electrons in Na atom = 11
No. of protons in Na atom = 11
Na atom loses one electron to become Na^+ ion. Where number of protons remains same. As the extra proton makes the net charge +1.

2. Correct option: (C) 18, 19, 20

The atomic number of potassium is 19. Since atomic number = number of protons = number of electrons.
Potassium ion has 18 electrons and 19 protons.
Mass number of potassium ion is 39.
We know, Number of neutrons = Mass number – number of protons
Hence, number of neutrons = 39 – 19 = 20

3. Correct option: (A) 7, 7, 7

The atomic number of Nitrogen is 7. Since atomic number = number of protons = number of electrons.
Nitrogen has 7 electrons and 7 protons.
Mass number of Nitrogen is 14.
We know, Number of neutrons = Mass number – number of protons
Hence, number of neutrons = 14 – 7 = 7

4. Correct option: (C) 9

The atomic number of Fluorine is 9. Since atomic number = number of protons

5. Correct option: (B) 31

Nucleons are number of neutrons and protons present in the nucleus of an atom.
Atomic number = number of protons = 15
Number of neutrons = 16
Hence, number of nucleons present in Phosphorus atom = (16+15) = 31

6. Correct option: (A) Hydrogen

Hydrogen atom doesn't contain any neutron. It has total one electron as it's atomic number is 1. The number of nucleon in H atom is 1 as it has only one proton.
Number of nucleons in He = 4 (2 protons + 2 neutrons)
Number of nucleons in Li = 7 (3 protons + 4 neutrons)
Number of nucleons in Be = 9 (4 protons + 5 neutrons)

7. Correct option: C) $_2He^4$

Nuclide is a species where the number of protons and neutrons is specific.
$_2He^4$ has 2 protons and 2 neutrons. So 2He4 is the Nuclide.
But $_6C^{10}$ has 6 protons and 4 neutrons.

$_{2}He^{8}$ has 2 protons and 6 neutrons.
$_{15}P^{29}$ has 15 protons and 14 neutrons.

8. Correct option: (A) Zero

neutrons are present inside nucleus. Proton is also a sub atomic particle like neutron

9. Correct option: (B) Carbon

Hydrogen doesn't have any neutron.
Carbon has 6 protons and 6 neutrons. So, the ratio of proton to neutron in carbon is 1:1.
Beryllium has 4 protons and 5 neutrons. So, the ratio is 4:5.
Sodium has 11 protons and 12 neutrons. So, the ratio is 11:12

10. Correct option: (D) Sodium

Carbon has 6 protons and 6 neutrons. So, the (p/n) ratio in carbon is 1:1
Nitrogen has 7 protons and 7 neutrons. So, the (p/n) ratio in nitrogen is 1:1
Oxygen has 8 protons and 8 neutrons. So, the (p/n) ratio in oxygen is 1:1
Sodium has 11 protons and 12 neutrons. So, the (p/n) ratio in sodium is 11:12

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct option: (C) All atoms have isotopes,

D) Atoms of different elements are different in all respects.
Modern atomic theory –
Atoms are no longer indivisible and consist of electrons, protons, neutrons and even more sub-particles.
Atoms of the same element may differ from one another called isotopes.
Atoms of different elements may be similar called isobars.
Atoms combine in a ratio which is not a simple whole number ratio, e.g. in sugar, the $C_{12}H_{22}O_{11}$ ratio is not a whole number ratio.

12. Correct options: A) Matter consists of small indivisible particles called atoms, B) Atoms of same element are like in all respects, C) Atoms combines in small whole numbers to form compound, D) Atom is the smallest unit of matter which takes part in a chemical reaction.

Matter consists of small indivisible particles called atoms.
Atoms of the same element are similar in every respect.
Atoms combine in a simple whole number ratio to form molecules.
Atoms of different elements are different.
Atoms can neither be created nor be destroyed.
Atoms can be rearranged, combined, or separated in chemical reactions.

13. Correct option: (A) $_{6}C^{12}$ B) $_{6}C^{13}$ C) $_{7}N^{14}$ D) $_{7}N^{15}$

All the elements here are neutral and doesn't carry extra electrons or protons. So (e/p) ratio is same in all elements.

14. Correct option: (A) C^{4-} B) N^{3-} C) O^{2-}

Atomic number of C = 4
Electrons in C^{4-} = 8
Atomic number of N = 5
Electrons in N^{3-} = 8
Atomic number of O = 6
Electrons in $^{O2-}$ = 8
Atomic number of Ne = 10
Electrons in Ne = 10

Assertion and reason type:

15. Correct Answer: A) Both A and R are true, and R is the correct explanation of A.

Explanation: For sodium (Na):

- Atomic number (Z) = 11
- Mass number (A) = 23 (common isotope)

Neutrons = A − Z = 23 − 11 = **12**

16. Correct Answer:B) Both A and R are true, but R is not the correct explanation of A.

Explanation: Dalton's theory → atoms are **indivisible** (True at that time)
Later discoveries (electron, proton, neutron) → atoms are **divisible**

Match the following:

17. Correct Matching is: a) → R), b) → S), c) → P), d) → Q)

number of neutrons are nearly same as the number of protons in an atom.
Atom with atomic number 7 has mass number of 14.
Atom with atomic number 9 has mass number of 19.
Atom with atomic number 12 has mass number of 24.
Atom with atomic number 18 has mass number of 40.

Comprehension type:

18. Correct Answer:C) Maharshi Kanada

This idea comes from ancient Indian philosophy.
It was proposed by **Maharshi Kanada**, who described matter as made of tiny indivisible particles called *paramanu*.

19. Correct Answer:B) Indivisible particle

"Paramanu" refers to the **smallest indivisible particle of matter**.

20. Correct Answer:B) Are the smallest particles that participate in chemical reactions

INTEGER TYPE

21. Correct Answer is: 1

Hydride ion (H^-) is the anion of hydrogen. It has an additional electron but number of protons is same as hydrogen.

22. Correct Answer is: 2

Hydrogen doesn't contain any neutron. It only carries one electron and one proton. So, the number of subatomic particles in hydrogen is 2.

23. Correct Answer is: 2

Atomic number of Calcium atom is 20, Mass number is 40. So the number of neutrons in calcium atom is 20.
10n = 20
N = 20/10 = 2

ISOTOPES, ISOTONES, AND ISOBARS
LEVEL-1

1. Correct option: (A)

Isotopes are the atoms of same element, having different mass number (same atomic number and different neutron number). In the given examples option (A) has the different atoms of same element hydrogen.It contains same atomic number but differ in their mass number because of different neutrons that are present in the nuclei.

2. Correct option: (C)

Isotopes are the atoms of same element having different masses. By having different number of neutrons in their nuclei they get these different masses. But they have same number of protons and electrons. The radiochemist Frederick Soddy was the one who discoved the Isotopes. In Greek Iso means same and Tope means place.

3. Correct option: (A)

Isotopes are the atoms of the same element having same atomic number because of this they will show same chemical properties. Isotopes have different mass number because of difference in the neutrons present in the nuclei, which is a physical property.

4. Correct option: (C)

The radiochemist Frederick Soddy was the one who discoved the Isotopes. In Greek Iso means same and Tope means place. Isotopes are the atoms of the same element having same atomic number and having different mass numbers.

5. Correct option: (C)

In the periodic table hydrogen is the smallest element.In Isotopes the atomic number is same so the all atoms of hydrogen isotopes occupies same place in the periodic table. The isotopes of Hydrogen are Protium, deuterium and Tritium. These isotopes have different mass number because of different number of neutrons present in them. Protium is the lightest isotope compared to the remaining because it does not contain neutrons.

6. Correct option: (C)

Many elements have non integral masses. This is because they have isotopes. We already know that isotopes are the atoms of same element having same atomic number but contain different mass. While giving the mass of an element having isotopes, we will take the average mass of all the isotopes. So many elements have non integral masses.

7. Correct option: (D)

Deuterium is the isotope of hydrogen, which is having same atomic number and different mass number. The isotopes are Protium, deuterium and tritium. The mass numbers of these three isotopes are 1,2 and 3 respectively. They have different mass numbers because of the presence of neutrons. So the number of neutrons present in deuterium is=A-Z=2-1=1

8. Correct option: (B)

We know that Isotopes are the atoms of same element having same atomic number and different mass numbers. In the given examples option(B) will satisfy the condition of Isotopes.

9. Correct option: (A)

The isotopes of hydrogen are Protium, deuterium and tritium. The mass numbers of these three isotopes are 1,2 and 3 respectively. In the given Isotopes the atomic number is same for all the Isotopes. Protium is the isotope which contains only one proton and no neutrons in its nucleous. So that isotope is also known as Protium.

10. Correct option: (B)

The atoms of different elements with same mass number but different atomic number are known as Isobars.

11. Correct option: (A)

Isobars are the atoms of different elements having same mass number but different atomic number. In the given options, option (A) contains atoms of different elements

(carbon and Nitrogen) having same mass number 14 for both the atoms but different atomic numbers 6 and 7 respectively.

12. Correct option: (A)

Isotones are the atoms of different elements having different atomic number and mass number but having same number of neutrons. we can find the number of neutrons by using formula

Number of neutrons (N)= mass number(A)-atomic number(Z)

=A-Z

For A) $_6C^{14,}$ $_7N^{15}$

Number of neutrons in carbon (N)=14-6=8

Number of neutrons in Nitrogen (N)=15-7=8

B) $_8O^{16,}$$_9F^{19}$

Number of neutrons in Oxygen (N)=16-8=8

Number of neutrons in Flourine (N)=9-19=10

C)$_1H^{1,}$ $_1H^2$

Number of neutrons in Protium (N)=1-1=0

Number of neutrons in Deuterium (N)=2-1=1

D)$_6C^{12,}$ $_6C^{13}$

Number of neutrons in carbon (N)=12-6=6

Number of neutrons in carbon (N)=13-6=7

In the given options, the elements of option (A) contain same number of neutrons, which satisfy the condition for Isotones.

13. Correct option: (C)

Among $_{10}A^{20}$,$_{11}B^{21}$,$_{12}C^{22}$,$_{11}D^{22}$

$_{12}C^{22}$,$_{11}D^{22}$ are Isobars. We know that isobars are those, the elements of different atoms having same mass number and having different atomic numbers. In these two elements are having same mass number but different in their atomic number and atoms also belongs to different elements.

14. Correct option: (B)

According to definition of Isobars, Isobars are the atoms of different elements having same mass number and having different atomic numbers.

15. Correct option: (C)

Isotones are the atoms of different elements having different mass number and different atomic number but having same number of neutrons.

We know, number of neutrons (N) = A-Z

Where A is mass number and Z is atomic number

In the given triads,

A) All atoms belong to same element, So these are not Isotones.

B) In the given atom Cu details were wrong

C) These are atoms of different elements

Number of neutrons in Ar=40-18=22

Number of neutrons in Ca=42-20=22

Number of neutrons in Sc =43-21=22

Numbers of neutrons in all the atoms are same and these are having different atomic number and different mass numbers

D) The atomic number for the atom F is wrong.

16. Correct option: (A)

The pair of atoms which belongs to the atoms of different elements having different atomic number and different mass number but having same number of neutrons.

A) Number of neutrons in K=mass number- atomic number=41-19=22
Number of neutrons in Sc=mass number-atomic number=43-21=22
So the given pair belongs to Isotones.

B) Not belongs to Isotones because the atoms belong to same element.

C) The number of neutrons are not same(22 and 19)

D) The number of neutrons are not same(20 and 22)

17. Correct option: (C)

The given pair contains same mass number and different atomic numbers and these atoms belong to different atoms, which satisfy the requirement of isobars. So the given atoms are isobars.

18. Correct option: (B)

The atom $_{33}As^{77}$ is the isotone of$_{32}Ge^{76}$, because these two belongs to atoms of different elements having different atomic number and mass number but having same number of neutrons.

19. Correct option: (A)

Let us calculate the number of neutrons present in all traids,

A) Number of neutrons in C=14-6=8
Number of neutrons in N=15-7=8
Number of neutrons in F=17-9=8

B) Number of neutrons in C=12-6=6
Number of neutrons in N=14-7=7
Number of neutrons in F=19-9=10

C) Number of neutrons in C=14-6=8
Number of neutrons in N=14-7=7
Number of neutrons in F=17-9=8

D) Number of neutrons in C=14-6=8
Number of neutrons in N=14-7=7
Number of neutrons in F=19-9=10

The given atoms in Option (A) belong to different atoms having different mass number and atomic number but having same number of neutrons belongs to isotonic triad.

20. Correct option: (C)

In Option (C) the given atoms belongs to different elements having same mass number but different atomic numbers, which is requirement for isobars. So these atoms will come under isobars.

21. Correct option: (B)

The sum of protons and neutrons is nothing but its mass number. The mass numbers are same in the elements they are known as Isobars.

22. Correct option: (C)

The elements given in option(C) represent isobars, because these are atoms of different elements having same mass number but different atomic numbers.

23. Correct option: (C)

Isoelectronic species are those, which are having same number of electrons.

A) The number of electrons is 14,14,17,14 respectively

B) The number of electrons is 14,14,14,15 respectively

C) The number of electrons is 14,14,14,14 respectively
D) The number of electrons is 15,14,14,17 respectively
So the spices given in option(C) represent isoelectronic species.

24. Correct option: (C)
In the given two nuclides the numbers of neutrons present are same. So these two are Isotones. This is the relation present between the two nuclides.

LEVEL-II

1. Correct option: (C)
Isotopes are the atoms of the same element having same atomic number but different mass numbers.
Mass number=atomic number + number of neutrons
Mass number of A=Y
Mass number of B=Y+1
As atomic number for both atoms is same(x), the difference between the number neutrons is nothing but difference between the mass numbers of the two atoms.
So difference between number of neutrons=Y+1-Y=1

2. Correct option: (B)
Isotopes are the atoms of the same element having same atomic number but different mass numbers.
Mass number=atomic number + number of neutrons
So in Isotopes as the atomic number same, the difference in the mass number is due to the presence of neutrons in their nuclei.

3. Correct option: (A)
The atoms of the elements represents Isotopes have similar chemical properties because of same number of electrons (atomic number).but they show different physical properties such as mass number.

4. Correct option: (C)
Given
Mass number=14
Number of neutrons in that atom=8
Atomic number (Z) = mass number-number of neutrons=14-8=6
So from the periodic table, We can identify that the element with atomic number 6 is carbon. So the given atom is isotope of Carbon.

5. Correct option: (B)
In the given
A) Isotones are atoms of different elements. But the given atoms are same elements So these are not isotones.
B) The given atoms represent isotopes, because these are atoms of same element having same atomic number and having different mass numbers.
C) The given atoms does not represents isobars, because isobars are atoms of different atoms having same mass number and different atomic number(mass number is not same)
D) Isotones are atoms of different elements, having different atomic number and mass number but having same number of neutrons. In the given elements the number of neutrons is not same.

6. Correct options: (B, C)

In the given options, Option (B) and (C) are not correct.

(B) Isotopes are the atoms of same element having same atomic number and different mass number.

(C) Isobars are atoms of different elements having same mass number and different atomic number

7. Correct options: (C, D)

In the given options,(C),(D) are not correct options.

(C) Both have different neutrons.In $_{17}Cl^{35}$ it has 18 neutrons and $_{17}Cl^{37}$ contains 20 neutrons.

(D) As the atomic number is same they have same chemical properties but not physical properties. They will show different physical properties such as having different mass number.

8. Correct options: (A, B)

Having same number of electrons, that elements are known as isoelectronic species.

A) Atomic number of Na is 11 as it lose one electron it forms Na+ion,No. of electrons in Na+=10

Atomic number of Mg is 12 as it loose 2 electrons it forms $Mg^{+2,}$ No. of electrons in Mg^{+2} =10

B) Atomic number of Mg is 12 as it loose 2 electrons it forms $Mg^{+2,}$ No. of electrons in Mg^{+2} =10

Atomic number of Al is 13 as it loose 3 electrons it forms $Al^{+3,}$ No. of electrons in Al^{+3}=10

C) Atomic number of O is 8 as it gains 2 electrons it forms $O^{-2,}$ No. of electrons in O^{-2} =10

Atomic number of N is 7, No. of electrons in N =7

D) Atomic number of Cl is 17 as it gain one electrons it forms $Cl^{-1,}$ No. of electrons in Cl^{-1} =18

Atomic number of Cl is 17, No. of electrons in Cl =17

So the number of electrons present in elements of Option (A) and (B) are same.

9. Correct option: (A, B, C)

The number of protons or electrons present in an atom is equal to atomic number of the particular atom.

A) Atomic number of Ca is 20m,So number of protons in Ca is 20

B) Atomic number of oxygen is 8, so number of electrons in Oxygen equal to 8.

C) Atomic number of carbon is 6, no. of electrons in C is 6.

D) Atomic number of Hydrogen is 1, so it contains only one protons but wrongly given as 12 protons present in Hydrogen.

10. Correct options: (B, C)

A) it is experimentally proved that neutron has no charge. it as neutral charge and it is denoted by zero.

B) Neutron present inside of the nucleus along with protons, combinedly known as nucleons. Neutron does not present outside of the nucleus.

C) Neutron mass is not same as that of electron. It is slightly more than proton.

D) Neutron is the heaviest particle in the atom, which is experimentally proved.

So the incorrect statements are B,C

11. Correct options: (B, C)

We know that Isotones are the atoms of different elements having different atomic number and mass number but have same number of neutrons.

A) No. of neutrons in $_{16}S^{30}$ =30-16=14
No. of neutrons in $_{15}P^{31}$=31-15=16
So these are not Isotones.

B) No. of neutrons in $_{9}F^{19}$=19-9=10
No. of neutrons in $_{10}Ne^{20}$ =20-10=10
These elements are Isotones.

C) No. of neutrons present in $_{19}K^{39}$=39-19=20
No. of neutrons in $_{20}Ca^{40}$ =40-20=20
These are Isotones

D) No. of neutrons in $_{11}Na^{23}$=23-11=12
No. of neutrons in $_{13}Al^{27}$=27-13=14
So option (B) and (C) have Isotones.

12. Correct options: (B, D)

We know that, atoms having same number of electrons are known as isoelectronic species.

A) Number of electrons in O^{-4}=8+4=12
Number of electrons in N^{-3}=7+3=10

B) Number of electrons in C^{-4}=6+4=10
Number of electrons in O^{-2}=8+2=10
These two species are isoelectronic species.

C) Number of electrons in O^{4+}=8-4=4
Number of electrons in Na^{+}=11-1=10

D) Number of electrons in Mg^{+2}=12-2=10
Number of electrons in Na^{+}=11-1=10
These are isoelectronic species.

13. Correct options: (A, B, C)

Isotopes are the atoms belong to same element having same atomic number but different mass number.

A) Given atoms are atoms of same element Hydrogen having same atomic number and having different mass number. So these are isotopes.

B) Given atoms are atoms of same element Carbon having same atomic number and having different mass number. So these are isotopes.

C) Given atoms are atoms of same element Oxygen having same atomic number and having different mass number. So these are isotopes.

D) The given atoms are atoms of different elements having different atomic number and mass numbers. So these are not isotopes.

So in the given options,(A,B,C) are isotopes.

14. Correct options: (A, B, C)

Isobars are those atoms of different elements having same mass number but different atomic numbers.

A) $_{6}C^{14}$,$_{7}N^{14}$ are the atoms of different elements having different atomic numbers and same numbers. So these are Isobars.

B) $_{8}O^{16}$,$_{7}N^{16}$ are the atoms of different elements having different atomic numbers and same numbers. So these are Isobars.

C) $_9F^{18}$,$_8O^{18}$ are the atoms of different elements having different atomic numbers and same numbers. So these are Isobars.

D) $_8O^{17}$,$_7N^{16}$ are the atoms of different atoms having different atomic number and different mass numbers. So these are not isobars.

Assertion and reason type:

15. Correct Answer: A) Both A and R are true, and R is the correct explanation of A.

Explanation: Isoelectronic species → same number of electrons

Example: O^{2-}, F^-, Na^+ (all have 10 electrons) But their nuclear charge (number of protons) is different This affects how strongly electrons are held → changes chemical behavior

16. D) A is false, R is true

Explanation: Isotones → same number of neutrons, **different atomic numbers**

Chemical properties depend on **electrons (atomic number)**, not neutrons

So isotones behave like **different elements**

Match the following:

17. Correct Matching is: a-S,b-R,c-Q.d-P

a)$_8O^{16}$,$_8O^{16}$ given in (S) are isotopes, because these two are atoms of same element having same atomic number but different in their mass number.

b) Isodiapheres are a set of nuclides with the same difference in the number of protons and neutrons in their nucleous.Both nuclides have have different number of protons and neutronsin their nuclei,but the difference is the same. In $_7N^{14}$ and$_8O^{16}$ difference in number of protons is1 and neutrons is 1 and same for both the atoms..So these are Isodiapheres

c) $_{14}$ Si^{30},$_{15}P^{31}$ are isotones because these are atoms of different elements having different atomic number and mass number but having same number of neutrons.

d) $_{18}Ar^{40}$$_{19}K^{40}$ are Isobars because these atoms are the atoms of different elements having different atomic number but having same mass number.

Comprehension type:

18. Correct Answer:B) 30

Explanation: Given:

Electrons = 26 → so protons = 26

Mass number (A) = 56

Neutrons = A − Z = 56 − 26 = 30

19. Correct Answer: C) 20 protons and 20 neutrons

In the nucleus:

- Protons = 20
- Neutrons = 40 − 20 = 20
- Electrons are not present in the nucleus

20. Correct Answer: C) 11 protons, 12 neutrons, 11 electrons

Explanation:

Given:

Mass number (A) = 23

Atomic number (Z) = 11

- Protons = 11
- Electrons = 11 (neutral atom)

- Neutrons = 23 − 11 = 12

21. Correct Answer is: 2

Tritium $_1H^3$ is the radioactive isotope of hydrogen. The number of neutrons present in tritium is=mass number – atomic number=3-1=2.

22. Atomic number of carbon is 6

BASIC LAWS IN CHEMISTRY

LEVEL – I

1. Correct option: (C) Conservation of mass

Chemical equation is balanced according to law of conservation of mass which states that mass in an isolated system neither created or destroyed by chemical reactions.

2. Correct option: (A) Lavoisier

French scientist, Lavoisier, in 1789 given the law of conservation of mass, which states that mass in an isolated system neither created or destroyed by chemical reactions.

3. Correct option: (C) Law of conservation of mass

Law of conservation of mass states that mass in an isolated system neither created or destroyed by chemical reactions. So the mass of the reactant is equal to the mass of the product. This is the law behind balancing of equations.

4. Correct option: (D) 28 parts of Nitrogen and 80 parts of oxygen

Number of moles of nitrogen = 2 mol
Atomic mass of nitrogen = 14 g/mol
Hence weight of nitrogen in N_2O_5 = 14×2 = 28 g
Number of moles of oxygen = 5 mol
Atomic mass of oxygen = 16 g/mol
Hence weight of oxygen in N_2O^5 = 16 × 5= 80 g
So part of nitrogen is 28 and oxygen is 80

5. Correct option: (D) A chemical compound consists of the same element combined together in the proportion by mass

Law of conservation of mass states that mass in an isolated system neither created or destroyed by chemical reactions. So the mass of the reactant is equal to the mass of the product. A chemical compound consists of the same element combined together in the proportion by mass

6. Correct option: (D) The law of conservation of mass

Law of conservation of mass states that mass in an isolated system neither created or destroyed by chemical reactions. So the mass of the reactant is equal to the mass of the product. This is the law behind balancing of equations.
Hence unbalanced chemical equation is against the law of conservation of mass.

7. Correct option: (C) Nuclear reactions

The law of conservation of mass is not acceptable for nuclear reaction, as it involves the conversion of energy from mass.

8. Correct option: (A) CO and CO_2

Solution: Law of multiple proportions states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers.
In CO and CO_2. O combines in a ratio 1:2 which is whole number. So it illustrates law of multiple proportions.

9. Correct option: (B) Proust

French chemist Joseph Proust proposed the law of constant proportion, which states that a chemical compound always contains exactly the same proportion of elements by mass.

10. Correct option: (C) Dalton

John Dalton proposed law of multiple proportions, which states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers.

11. Correct option: (A) Law of constant proportions

Percentage of carbon = 27.27%
Percentage of oxygen = 72.73%
Carbon : oxygen = 27.27/72.73 = 3:8
In CO_2, carbon : oxygen = 3:8
So it follows law of constant proportion.

12. Correct option: (D)$SnCl_2$,$SnCl_4$

Law of multiple proportions states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers.
In $SnCl_2$ and $SnCl_4$, Cl combines in a ratio, 2:4 = 1:2
So it illustrates law of multiple proportions.

13. Correct option: (A) Law of multiple proportions

We can clearly see that weight of carbon in both cases are same. But the weight of oxygen is different.
Law of multiple proportions states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers.
The ratio of oxygen in both compounds is, 16:32 = 1:2.
So this obeys law of multiple proportions.

14. Correct option: (C) 1:2

Relative mass of one H atom is 1.
Relative mass of one O atom is 16.
So their ratio of combining = 1:2

15. Correct option: (B)H_2O,H_2O_2

$2H_2 + O_2 \rightarrow H_2O$
$O_2^{-\cdot} + 2H^+ + e^- \rightarrow H_2O2$
$HCHO + OH \rightarrow HO_2$
$H_2O + H^+ \rightarrow H_3O^+$

16. Correct option: (D) Multiple proportion

Law of multiple proportions states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers.

17. Correct option: (A) Constant proportions

Law of constant proportion states that a chemical compound always contains exactly the same proportion of elements by mass.
Percentage of copper and oxygen in samples of CuO obtained by different methods were found to be the same. This proves law of constant proportion as Cu : O remains constant.

18. Correct option: (D) Sulphur dioxide, sulphur trioxide

Law of multiple proportions states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers.
In case of Sodium chloride, sodium bromide &
Ordinary water, heavy water & Caustic soda, Caustic potash different types of ions are associated. So law of multiple proportions is not applicable.
Sulphur dioxide has 2 oxygen. Sulphur trioxide has 3 oxygen. The ratio is 2:3.
Hence it obeys law of multiple proportions.

19. Correct option: (A) Law of constant proportions

According to the law of constant proportion in a given compound, the elements always combine in the same proportion as each other. In H2O the ratio of H : O = 1 : 8. Which remains constant.

20. Correct option: (A) Law of constant proportions

Law of constant proportion states that a chemical compound always contains exactly the same proportion of elements by mass.
Percentage of copper and oxygen in samples of CuO obtained by different methods were found to be the same. This proves law of constant proportion as Cu : O remains constant.

LEVEL - II

1. Correct option: (D) CO, CO_2

Law of multiple proportions states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers.
In CO and CO_2. O combines in a ratio 1:2 which is whole number. So it illustrates law of multiple proportions.

2. Correct option: (B) H_2O, D_2O_2

Law of multiple proportions states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers
It is applicable for compound made with same elements.
As H_2O and D_2O_2 are made of different elements. Thus it doesn't illustrate law of multiple proportions.

3. Correct option: (B) Law of multiple proportion

Law of multiple proportions states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers
Compound A = 1.00 g nitrogen + 0.57 g Oxygen
Compound B = 2.00 g nitrogen + 2.24 g oxygen
Compound C = 3.00 g nitrogen + 5.11 oxygen
So If we fixed the amount of nitrogen (1 g) for all compound then, For compound A
1.00 g of N combine with 0.57 g of O_2
For compound B :
2 g N of N combine with 2.24 g of O_2
1g N = 1.12g O_2
For compound C
3 g N combine with 5.11 O_2
1 g N = 1.70 g O_2

Ratio of oxygen = 1 : 2 : 3

4. Correct option: (C) Law of multiple proportion

Law of multiple proportions states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers.

We can see that, the weight of lead combining with fixed weight of oxygen are in the ratio of 1:2. Which illustrates law of multiple proportion.

5. Correct option: (B) n + m = p + q

Law of conservation of mass states that mass in an isolated system neither created or destroyed by chemical reactions. So the mass of the reactant is equal to the mass of the product.

According to law of conservation of mass

Total weight of reactants = Total mass of products

→ n + m = p + g

6. Correct option: (A) Law of multiple proportion

Law of multiple proportions states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers

SO_2 has 2 oxygen. SO_3 has 3 oxygen. The ratio is 2:3. Hence it obeys law of multiple proportions.

7. Correct option: (A) Law of multiple proportion

Law of multiple proportions states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers

ONE OR MORE THAN ONE OPTION CORRECT TYPE

8. Correct option: (A) Matter can be neither created nor destroyed, B) Total mass of reactants is equal to total mass of products, D) A chemical compound consists of the same element combined together in the same proportion by mass

Law of conservation of mass states that mass in an isolated system neither created or destroyed by chemical reactions. So the mass of the reactant is equal to the mass of the product. It also states that matter can neither be created nor destroyed. Also a chemical compound consists of the same element combined together in the proportion by mass

9. Correct option: (A) N_2O,NO,B) NO_2,N_2O_4, D) N_2O, NO_2, N_2O_5

Law of multiple proportions states that when two elements combine together to form more than one compound, the weight of one element that combine with a fixed weight of the other in a ratio of small whole numbers

Hence, for a set of compounds to follow law of multiple proportions, the combining elements must be the same.

N_2O_5 and P_2O_5 don't follow the law of multiple proportions.

For N_2O and NO, NO_2 and N_2O_4,N_2O, NO_2, N_2O_5.

Oxygen combines in a fixed ratio.

10. Correct option: (A) 1:2

In carbon monoxide 12 parts carbon combine with 16 parts of oxygen.

In carbon dioxide 12 parts of carbon combine with 32 parts of oxygen.

The ratio of masses of oxygen is 16:32 or 1:2

11. Correct option: (A) Law of conservation of mass, B) Law of constant proportions, C) Law of multiple proportions

Law of conservation of mass,Law of constant proportions & Law of multiple proportions are law of chemical combinations.

Assertion and reason type:

12. Correct Answer:

A) Both A and R are true and R is correct explanation

This is based on the Law of Conservation of Mass.

It states that mass remains constant during a chemical reaction**.**

13. Correct Answer: C) A is true, R is false

Assertion is based on the Law of Definite Proportions

→ A compound always has the same composition by mass

Reason says composition depends on preparation → incorrect

14. Correct Matching is: (a) →Q), b) →S), c) →P), d) → R)

French scientist, Lavoisier, in 1789 given the law of conservation of mass, which states that mass in an isolated system neither created or destroyed by chemical reactions.

Law of multiple proportions was given by Dalton in1803 and law of constant proportion was given by Proust in 1797.

Mass of hydrogen = 4

Mass of oxygen = 32

So the ratio of mass of hydrogen and oxygen = 4:32 = 1:8

Comprehension type:

15. Correct option: (D) It violates law of conservation of mass.

Law of conservation of mass states that mass in an isolated system neither created or destroyed by chemical reactions. So the mass of the reactant is equal to the mass of the product.

Here 100 grams of reactants are reacted to give 90 grams of products.

So it violates law of conservation of mass

16. Correct option: (C) 44g of CO2

Solution: $C + O_2 \rightarrow CO_2$

According to law of conservation of mass, mass of the reactant is equal to the mass of the product.

C = 12 g

O_2 = 32 g

Hence CO_2 = 32+12 = 44 g

17. Correct option: (B) 32g

According to law of conservation of mass, mass of the reactant is equal to the mass of the product.

$2H_2 + O_2 \rightarrow 2H_2O$

4 + x = 36

x = 32

Integer type:

18. Answer: Mass of CO_2 = 44 g

Given:

- $CaCO_3$ (reactant) = 100 g
- CaO (product) = 56 g

Reaction:
$CaCO_3 \rightarrow CaO + CO_2$
Let mass of CO_2 = x
Using conservation of mass:
Total reactant mass = Total product mass
100 = 56 + x
x = 100 − 56 = 44 g
19. Answer:0 (zero)
Explanation:

- Mass of reactants = Mass of products
- So, no change in total mass

Chemical Equations

LEVEL-I

1. Correct option: (B) AgCl – White precipitate
$AgNO_3 + NaCl \rightarrow AgCl\downarrow + NaNO_3$
Colour of AgCl is white.

2. Correct option: (B) Chemical Decomposition
A decomposition reaction is a reaction in which a compound breaks down into two or more simpler substances.
Here Lead nitrate decomposes into three simple compounds. So it's a decomposition reaction.

3. Correct option: (B) $4Na + O_2 \rightarrow 2Na_2O$
$4Na + O_2 \rightarrow 2Na_2O$
Here two elements combine together to form a compound.
In other cases combination of compounds are involved.

4. Correct option: (B) Chemical Decomposition
A decomposition reaction is a reaction in which a compound breaks down into two or more simpler substances is known as chemical decomposition.
Here water decomposes into two simple elements. So it's a chemical decomposition.

5. Correct option: (A) Chemical Decomposition
A decomposition reaction is a reaction in which a compound breaks down into two or more simpler substances.
Here $NaNO_3$ decomposes into two simple substances. So it's a chemical decomposition.

6. Correct option: (A) Chemical Combination
A combination reaction in which a compound is formed by the combination of its Constituent element is known as chemical combination.

7. Correct option: (A) Chemical Combination
A combination reaction in which a compound is formed by the combination of its Constituent element is known as chemical combination.
Here CaO and H_2O combine together to form $Ca(OH)_2$. So it's a chemical combination.

8. Correct option: (A) Chemical Combination
A combination reaction in which a compound is formed by the combination of two or more elements or compounds is known as chemical combination.

9. Correct option: (A) Combination reaction
A combination reaction in which a compound is formed by the combination of two or more elements or compounds is known as chemical combination.

Here two elements Na and Cl_2 combine together to form NaCl. So it's a combination reaction.

10. Correct option: (D) $2Ag + Br_2 \rightarrow 2AgBr$

A decomposition reaction is a reaction in which a compound breaks down into two or more simpler substances.

HgO decomposes to Hg and O_2.

$CaCO_3$ decomposes to CaO and CO_2.

H_2O decomposes to H_2 and O_2.

But Ag and Br_2 combine together to form AgBr. So it's a combination reaction.

11. Correct option: (D) A double displacement reaction

A reaction in which two compounds exchange their ions to form two new compounds is known as double displacement reaction.

Displacement reaction is a reaction in which one kind of atom is replaced by the other kind of atom in the solution and results in a new product formation.

A decomposition reaction is a reaction in which a compound breaks down into two or more simpler substances.

An addition reaction is a reaction in which two or more compounds or elements combine together to form a new compound.

12. Correct option: (C) Chemical Double Displacement

A reaction in which two compounds exchange their ions to form two new compounds is known as chemical double displacement reaction.

So it is a chemical double displacement reaction.

13. Correct option: (B) $Ca + Cl_2 \rightarrow CaCl_2$

single displacement reaction is a reaction in which one kind of atom is replaced by the other kind of atom in the solution and results in a new product formation.

Here Ca and Cl_2 combine together to form $CaCl_2$. So it is a chemical combination reaction.

A combination reaction in which a compound is formed by the combination of two or more elements or compounds is known as chemical combination reaction.

14. Correct option: (D) Both 'A' and 'B'

A reaction in which two compounds exchange their ions to form two new compounds is known as chemical double displacement reaction.

Here Ag^+ changes with Na^+ Ion and Cl^- changes with NO_3^{2-} ion. So it is a double displacement reaction.

Again precipitation of AgCl is observed. So it also a precipitation reaction.

15. Correct option: (B) 1 and 2

Neutralization reaction is a reaction in which a base and a acid react together to form a neutral environment.

Here strong acid H_2SO_4 and strong base NaOH reacts together to create neutral environment. So it's a Neutralization reaction

A reaction In which two compounds exchange their ions to form two new compounds is known as chemical double displacement reaction.

Here H^+ changes with Na^+ ion and SO_4^{2-} changes with OH^- ion. So it is a double displacement reaction.

16. Correct option: (B) Single displacement reaction

single displacement reaction is a reaction in which one kind of atom is replaced by the other kind of atom in the solution and results in a new product formation.

Here Zn Is replaced with $ZnSO_4$.

So it is a single displacement reaction.

LEVEL – II

1. Correct option: (A) AB → A + B

A decomposition reaction is a reaction in which a compound breaks down into two or more simpler substances.

Here AB decomposes to simpler substances A & B.

In the other cases all elements combine together to form complex compounds. So rest of three are combination reaction.

2. Correct option: (A) $C_6H_{12}O_6$

Balancing the equation we get, the correct answer is $C_6H_{12}O_6$ (Glucose)

3. Correct option: (B) The number of atoms present in products side

Balancing a chemical equation obeys the law of conservation of mass.

A chemical equation is balanced when the number of atoms of an element in reactant Side is equal to the number of atoms present in products side.

Hence option B) is correct.

4. Correct option: (A) Ammonia

Nitrogen reacts with hydrogen gives ammonia (NH_3) H_2 (g) + N_2 (g) → NH_3 (g)

By Balancing the equation we get, $3H_2$ (g) + N_2(g) → $2NH_3$ (g)

This is known as the Haber process.

5. Correct option: (B) Carbon dioxide

$CaCO_3$ (s) _heat → CaO (s) + CO_2 (g)

6. Correct option: (D) White precipitate

Sodium sulphate (Na_2SO_4) and barium chloride ($BaCl_2$) reacts together in aqueous medium to form solid precipitate of barium sulphate ($BaSO_4$) and Sodium chloride (NaCl) is also formed. $BaCl_2$(aq) + Na_2SO_4 (aq) → $BaSO_4$ (s) ↓ + NaCl (aq)

7. Correct option: (B) Iron is more reactive than copper

Copper is less reactive than iron. Therefore displacement reaction doesn't occur and copper can not displace iron from iron sulphate solution.

8. Correct option: (A) $C + O_2 \rightarrow CO_2$

Non-stoichiometric reaction is a reaction in which law of constant proportion is violated.

9. Correct option: (A) CaO

The reaction is, $CaCO_3 \rightarrow CaO + CO_2 \uparrow$

As CO_2 is a gas, CaO remains as residue.

10. Correct option: (B) ↓

The precipitate is usually settle down in a mixture. Thus downward arrow is used for precipitate.

11. Correct option: A) Heat B) Light C) Electricity

A decomposition reaction is a reaction in which a compound breaks down into two or more simpler substances. The reaction is often initiated by the application of heat, light, electricity.

Whereas bacteria decompose dead tissue or organisms. They can't carry decomposition reaction.

12. Correct option: (C) O_2

Electrolysis of water

Anode – $2H_2O = O_2 + 4H^+ + 4e^-$

Cathode – $2H_2O + 2e^- = H^2 + 2OH^-$

So, electrolysis of water gives oxygen and hydrogen.

13. Correct option: (A) Neutralization reaction B) Double displacement reaction C) Acid base reaction

Neutralization reaction is a reaction in which a base and a acid react together to form a neutral environment.

Strong acid HCl and NaOH reacts together to make NaCl and H_2O neutral compounds.

A reaction in which two compounds exchange their ions to form two new compounds is known as chemical double displacement reaction.

The Ions are exchanging their positions. So it is a double neutralization reaction.

A reaction that occurs between a base and an acid is known as acid base reaction.

Here HCl is acid and NaOH is base. So it is a acid base reaction.

Assertion and reason type:

14. Correct Answer:

A) Both A and R are true and R is correct explanation

First balance the equation:

Correct balanced form is:

$2H_2 + O_2 \rightarrow 2H_2O$

Now check atoms:

- Left side: H = 4, O = 2
- Right side: H = 4, O = 2

So atoms are equal on both sides.

15. Correct Answer:

A) Both A and R are true and R is correct explanation

In photosynthesis, plants convert:

Carbon dioxide + Water → Glucose + Oxygen

New substances are formed → this is a chemical reaction

Match the following:

16. Correct Matching is: (a) →R), b) →S), c) →P),d) →Q)

A reaction in which two compounds exchange their ions to form two new compounds is known as chemical double displacement reaction. Ag^+ and Na^+ ions changes their positions as well as NO_3^{2-} and Cl^- ions changes their positions.

A decomposition reaction is a reaction in which a compound breaks down into two or more simpler substances.

$MgCO_3$ decomposes to simpler compounds MgO and CO_2.

Chemical displacement reaction is a reaction in which one kind of atom is replaced by the other kind of atom in the solution and results in a new product formation.

Na is replaced by NaOH

A chemical combination reaction in which a compound is formed by the combination of two or more elements or compounds CaO and H_2O combine together to form $Ca(OH)_2$.

Comprehension type:

17. Correct Answer:B) Chemical reaction

A chemical equation is a symbolic way of showing a chemical reaction, using symbols and formulas of reactants and products.

18. Correct Answer: B) To follow conservation of mass

A chemical equation must follow the Law of Conservation of Mass, which states that mass is neither created nor destroyed.

19. Correct Answer:C) Equal on both sides

A balanced equation means:

Same number of each type of atom on reactant and product side

Example:

$2H_2 + O_2 \rightarrow 2H_2O$

H = 4 on both sides, O = 2 on both sides

20. Correct Answer is: equal

one molecule $CaCO_3$ will decompose to one molecule CaO and one molecule CO_2. So it will give equal number of CO_2 molecules.

21. Correct Answer is: 6

molecular formula of glucose → $C_6H_{12}O_6$.

So number of oxygen atoms → 6

22. Correct Answer is: 6

Balancing the equation we get X = 6

23. Correct Answer is: 2

Balancing the equation we get, X = 2

Previous year questions:

1. Proton numbers of elements X and Y are 6 and 7 respectively.

- X has 6 protons.
- Y has 7 protons.

Statement II is True.

III. Elements X and Y are represented by $^{14}_{6}X$ and $^{14}_{7}Y$ respectively.

- X → Atomic number = 6, Mass number = 14 → $^{14}_{6}X$
- Y → Atomic number = 7, Mass number = 14 → $^{14}_{7}Y$

Statement III is True.

IV. Elements X and Y are isobars.

Isobars are atoms having:

- Same mass number
- Different atomic numbers

Here,

- Mass numbers = 14 and 14 (same)
- Atomic numbers = 6 and 7 (different)

Therefore, X and Y are isobars.

Statement IV is True.

2.

Atom	Protons	Electrons	Neutrons
P	12	12	12
Q	12	12	14

Step 1: Calculate Atomic Number and Mass Number

- **Atomic Number (Z)** = Number of protons
- **Mass Number (A)** = Protons + Neutrons

For **P**

- Atomic number = 12
- Mass number = 12 + 12 = **24**

For **Q**

- Atomic number = 12
- Mass number = 12 + 14 = **26** so, C) P and Q are isotopes

3. **Statement II**

Increasing mass number:

35 < 39 < 40 < 80 < 127

So,

W < U < Y < X < V

Statement III

Order of proton numbers:
W < Y < U < X < V
Order of mass numbers:
W < U < Y < X < V
These two orders are **different**.

4. **Decomposition reaction** :Water breaks down into hydrogen and oxygen.

5. Step 1: Analyse the Assertion
An atom is electrically neutral because it has an equal number of protons and electrons.
Isotopes are atoms of the same element, so they also have equal numbers of protons and electrons. Therefore, isotopes are electrically neutral.
Assertion (A) is True.
Step 2: Analyse the Reason
The given definition is incorrect.
The correct definition of isotopes is:
Isotopes are atoms of the same element having the same atomic number but different mass numbers.
Examples:

- $^{35}_{17}$Cland $^{37}_{17}$Cl
- Both have atomic number = 17
- Mass numbers are 35 and 37

The reason states:

- Same mass number but different atomic number.

This is actually the definition of isobars, not isotopes.
Therefore,
Reason (R) is False.

5. Changes around Us

LEVEL – I

1. Correct option: (B)

Changes that occur within seconds or minutes are considered fast changes because they happen relatively quickly compared to slow changes, which might take hours, days, or longer to occur.

2. Correct option: (C)

The process of dry leaves decomposing and transforming into manure is a **slow change**. It takes a significant amount of time for the organic matter in the dry leaves to break down and convert into nutrient-rich compost or manure. This is a natural and beneficial process in agriculture and gardening.

3. Correct option: (A)

The process of evaporation is caused by heating. When a liquid is heated, its molecules gain energy and become more energetic, leading to increase in kinetic energy. As a result, some of these molecules gain enough energy to break their bonds and escape into the gaseous phase, causing evaporation. Evaporation is the phase transition from a liquid to a gas due to the input of heat energy.

4. Correct option: (D)

The blackening of silverware occurs due to a chemical reaction with sulfur compounds in the environment, is a slow change. It occurs gradually over time as the silver reacts with sulfur to form silver sulfide, tarnishing the surface. In contrast, the other options (burning of a matchstick, spinning of a top, and bursting a balloon) involve relatively faster processes or actions.

5. Correct option: (A)

A reversible change is one type of change that can return the system or object to its original state or condition. In other words, the changes that occur during a reversible change are temporary and can be reversed with the appropriate conditions or actions.

6. Correct option: (C & D)

Water changes into ice - This is a reversible change because you can melt the ice (solid water) back into liquid water by applying heat.

Blowing a balloon - This is also a reversible change. You can deflate the balloon to return it to its original state.

7. Correct option: (B)

An irreversible change is one type of change that cannot be undone, meaning the system or object cannot be returned to its original state or condition. Irreversible changes are permanent and involve significant alterations to the system or object.

8. Correct option: (C)

The formation of ash from burnt paper is an example of an irreversible change. Once the paper is burnt and turns into ash, it cannot be easily reversed or returned to its original form. Irreversible changes typically involve the transformation of substances into new forms that cannot be easily reversed.

9. Correct option: (D)

Evaporation is an example of both a reversible change and a physical change. It is reversible because you can condense the vapor back into a liquid by cooling the gas, and it is a physical change because the substance (liquid to gas) changes its state without undergoing a chemical reaction.

10. Correct option: (B)

In the boiling process, liquid water changes into vapor (water in the gaseous state). This is a physical change where the substance changes its state from a liquid to a gas due to the input of heat energy, without undergoing a chemical reaction.

11. Correct option: (C)

In the melting process, a solid substance changes into its liquid state. This is a phase transition where a substance gains enough heat energy to break the intermolecular forces holding its particles in a solid arrangement, allowing them to move more freely in a liquid state.

12. Correct option: (A)

Condensation is the process in which water vapor changes into liquid water. This occurs when the water vapor loses heat energy and transitions back to its liquid state, forming tiny water droplets.

13. Correct option: (C)

The undesirable change is "**Food turning bad in summer**." Food spoiling or turning bad is generally considered undesirable because it can lead to food borne illnesses and waste of food resources. The other options, such as the change of seasons, evaporation of water, and curdling of milk, are natural processes or reversible changes that are not necessarily undesirable in themselves.

14. Correct option: (A)

The melting of ice candy is a reversible change because you can easily change it back to a solid state by cooling it down. It involves a phase transition from a solid (ice candy) to a liquid (melted candy) due to the addition of heat, and this process can be reversed by removing the heat and allowing it to solidify again.

15. Correct option: (A)

Vaporization is a reversible change because it involves the transition of a substance from its liquid phase to its gaseous phase due to the addition of heat, but this change can be reversed by cooling the gas to turn it back into a liquid. It is not a permanent change because it can be undone.

16. Correct option: (B)

Curdling of milk is a non-periodic change. It is not a regular or recurring event like seasonal changes, for example. It can happen under specific conditions, such as when milk is exposed to certain bacteria or acids, leading to the separation of curds and whey. This change is not necessarily desirable but can be controlled in various culinary processes to make cheese and other dairy products. It is also considered a physical change because the chemical composition of milk remains the same; only its physical state changes as curds and whey form.

17. Correct option: (B)

Ozone depletion refers to the decrease in the concentration of ozone molecules in the Earth's ozone layer, particularly in the stratosphere. This depletion is caused by human-made chemicals known as ozone-depleting substances, such as **chlorofluorocarbons** (CFCs), which lead to the thinning of the ozone layer. This thinning of the ozone layer can have harmful effects, including increased exposure to harmful ultraviolet (UV) radiation from the sun.

18. Correct option: (A)

High and low tides in the ocean are periodic changes because they occur in a regular and predictable pattern over time. They are primarily driven by the gravitational forces of the Moon and the Sun and follow a cyclical pattern of rising

and falling water levels. These changes happen in a rhythmic, repeating manner, making them periodic in nature.

19. Correct option: (A)

The phases of the Moon are a periodic change. The Moon goes through a regular cycle of phases as it orbits the Earth, which includes new moon, first quarter, full moon, and last quarter phases, among others. These phases occur in a predictable and repetitive pattern, making them periodic in nature.

20. Correct option: (B)

An earthquake is a non-periodic change because it is not a regular or recurring event like the beating of a heart (which is periodic), the change of seasons (which is periodic based on the Earth's orbit around the Sun), or the swinging of a pendulum (which is periodic). Earthquakes occur sporadically and unpredictably, making them non-periodic events.

LEVEL – II

1. Correct option: (B)

The conversion of milk into curd is an irreversible change. Once milk undergoes the curdling process, it cannot be easily reverted back to its original state as milk. This change involves a chemical transformation due to the action of bacteria or enzymes, resulting in the formation of curd with a different composition and properties compared to milk.

2. Correct option: (C)

Drawing a picture on a drawing sheet with a pencil is an example of a reversible change. You can erase the pencil marks and return the drawing sheet to its original blank state. The marks made by the pencil are not permanent and can be easily undone, making this a reversible alteration to the drawing sheet.

3. Correct option: (B)

Burning of coal that produces a lot of smoke and other poisonous gases is generally considered an undesirable change because it has negative environmental and health consequences. It leads to air pollution and can harm human health and the environment.

4. Correct option: (A)

The heat produced by the burning of petrol in the engine of a car or scooter is a desirable change. This heat is harnessed to power the engine and drive the vehicle, making it a useful and necessary outcome of the combustion process. It's part of the intended functionality of the engine and serves a valuable purpose in transportation.

5. Correct option: (B)

Baking dough into bread is an irreversible change because the process involves chemical reactions, such as the Maillard reaction and the caramelization of sugars that transform the dough into bread with a different composition and properties. Once the dough is baked, it cannot be easily reverted back to its original raw dough form.63

6. Correct option: (A, C)

Melting of ice: This is a physical change because, ice (solid water) is still in a different physical state (liquid) and its chemical composition remains the same.

Heating of Nichrome wire: This is a physical change because it involves heating a solid metal (Nichrome wire) to make it hotter, but the chemical composition of the wire remains unchanged.

7. Correct option: (A)

Germination of a mustard seed into a seedling is a slow change. This statement is correct because germination is a gradual process that takes time for the seed to develop into a seedling.

8. Correct option: (D)

Changing of seasons is a slow change. This statement is true. The changing of seasons, such as from winter to spring or summer to fall, is a gradual and slow process that occurs over time as the Earth orbits the Sun, causing shifts in temperature and weather patterns.

9. Correct option: (C)

A change in which the products formed cannot be converted back into reactants.

10. Correct option: (A, B, D)

Germination of a mustard seed: Germination is a non-periodic change because it occurs once when a seed starts to grow into a plant, and it does not follow a regular repeating pattern.

Curdling of milk: Curdling of milk is a non-periodic change as it occurs when specific conditions (such as adding an acid or rennet) are met and is not a regular, repeating process.

Falling of leaves: It is a non-periodic change. It occurs when trees shed their leaves in response to seasonal changes, and the timing can vary from tree to tree and year to year. It is not a regularly repeating process.

11. Correct option: (A, B)

Formation of manure from animal dung: This is a desirable change because it involves the decomposition of organic matter (animal dung) into nutrient-rich manure, which can be used to improve soil fertility and support plant growth.

Formation of curd from milk: This is a desirable change because the formation of curd from milk is a controlled process that is intentionally carried out by people. Curd is a valuable dairy product with nutritional benefits and various culinary uses.

12. **Answer: (D**) Assertion is false, but Reason is true.

13. **Answer: (A)** Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

14. **Answer: (C)** Assertion is true, but Reason is false.

15. Correct Matching is: a-R; b-T; c-Q; d- P

a) Very fast change is bursting of balloon.
b) Reversible change a change can be reversed.
c) Irreversible change a change cannot be reversed.
d) Very slow change germination of mustard seed.

16. Correct Matching is: a-R; b-P; c-S; d- Q

a) Periodic change is change of seasons.
b) Slow change formation of biogas.
c) Desirable change baking of cake.
d) Reversible change melting of wax.

17. Correct option: (A)

Cooking of food: Cooking food is irreversible change. Once food is cooked, its chemical composition and physical properties may change, making it difficult or impossible to revert to its original raw state.

18. Correct option: (D)

Condensation of water: Condensation is a reversible change. When water vapor cools and condenses, it forms liquid water. This process can be reversed by heating the liquid water to convert it back into water vapor (evaporation).

19. Correct option: (B)

When food is digested in the human body, it undergoes chemical and physical transformations that break it down into simpler substances that can be absorbed and used by the body. These changes cannot be easily reversed to reconstruct the original food.

20. Correct option: (D)

Cooking of food is a desirable change because it improves the taste, texture, and digestibility of many food items, making them safer to consume and enhancing their nutritional value. Cooking kills harmful bacteria and parasites, making the food safer to eat, and it also makes certain nutrients more readily available for absorption by the body.

21. Correct option: (C)

Rusting of iron is an undesirable change because it leads to the corrosion and deterioration of iron objects, causing them to weaken and eventually become useless. This process can be harmful and costly when it occurs in structures, machinery, or equipment made of iron or steel. The other options, such as the change of seasons, melting of snow, and curdling of milk, do not generally have negative or harmful consequences.

22. Correct option: (A)

Ripening enhances the taste, texture, and aroma of fruits, making them more palatable and enjoyable to eat. It also increases the fruit's nutritional value in some cases. Therefore, ripening is usually viewed positively in the context of consuming fruits.

PHYSICAL AND CHEMICAL CHANGES

LEVEL - I

1. Correct option: (A)

Change is generally temporary in nature and no new substances are formed is called as physical change.

2. Correct option: (D)

Physical changes can be generally reversible and irreversible.

3. Correct option: (C)

On heating, the color of zinc oxide changes from white to yellow.

4. Correct option: (B)

Physical change can be easily reversed.

5. Correct option: (C)

In a physical change, energy is either absorbed or evolved.

6. Correct option: (C)

Boiling of water is an example of physical change.

7. Correct option: (B)

A physical change is temporary.

8. Correct option: (A)

In physical change composition remains the same.

9. Correct option: (A)

Physical change is generally temporary in nature and no new substances are formed.

10. Correct option: (A)

When ice is heated to 10^0C, it's state changes to liquid state.

11. Correct option: (B)

Evaporation is a physical change.

12. Correct option: (A)

The browning of banana in the air is a chemical change.

13. Correct option: (B)

Cooking of food is an example of chemical change.

14. Correct option: (A)

Chemical change cannot be easily reversed.

15. Correct option: (B)

A new substance is formed with different properties from the original substance then the change is chemical change.

16. Correct option: (C)

Clotting of blood is a chemical change.

17. Correct option: (B)

Chemical change in which new substances are formed.

18. Correct option: (D)

Burning of a match stick is a chemical change.

19. Correct option: (B)

On heating, sodium nitrate ($NaNO_3$) it undergoes a decomposition reaction and gives Sodium nitrite ($NaNO_2$) and Oxygen (O_2) as its product. It is a chemical change.

20. Correct option: (C)

In a chemical change, energy is either absorbed or evolved.

LEVEL - II

1. Correct option: (C)

$2Mg(s)+O_2(g)\rightarrow 2MgO(s)+heat$ Reaction illustrates a chemical change.

2. Correct option: (C)

Preparation of food material by plants is an example of chemical change. It is permanent and can't be reversible.

3. Correct option: (C)

Mass remains the same during physical and chemical change.

4. Correct option: (B)

Magnesium ribbon burns with brilliant white flame.

5. Correct option: (B)

Dry ice (solid) is sublimated at room temperature is a chemical change.

6. Correct option: (C)

When Zinc hydroxide is heated, it decomposes to form Zinc oxide. This is an example of chemical change.

7. Correct option: (B)

Rusting of iron is a chemical change.

8. Correct option: (A)

Dew is formed when water vapor condenses into water droplets. No change in chemical composition of water.

9. Correct option: (A)

Water boils at 100^0C. It is a physical change.

10. Correct option: (A)

It involves a change of state from solid to liquid. Melting of wax is a physical change.

11. Answer: A) Charcoal and **B) Steam**

- *Explanation:* When sugar ($C_{12}H_{22}O_{11}$) is heated strongly, it undergoes a dehydration reaction, leaving behind a black residue of carbon (sugar charcoal) and releasing water vapor (steam).

12. Answer: B) Permanent change (Chemical change)

- *Explanation:* Heating mercuric oxide decomposes it into mercury metal and oxygen gas. This completely alters the chemical composition, making it a permanent chemical change.

13. Answer: A) It is a physical change and **B) No new products are formed**

- *Explanation:* (Note: The question wording implies multiple options can be valid observations). Dissolving salt in water is a physical change because the salt can be recovered by evaporation, and its chemical identity does not change.

14. Answer: A, B, and C are all physical changes.

- *Explanation:* Formation of salt solution, magnetization of iron, and evaporation of water do not form new chemical substances. Only **D) Burning of Mg ribbon** is a chemical change. If you must choose only one absolute standard textbook example of a purely physical state/phase change, **C) Evaporation of water** is the most definitive.

15. Answer: A) Photosynthesis by plants and **B) Formation of wine**

- *Explanation:* Both processes involve chemical reactions that create entirely new substances. Melting of wax (C) is a physical phase change, and a rolling rock (D) is a mechanical/physical event.

Assertion and Reason Type

16. **Assertion (A):** In a physical change, no new substance is formed.
Reason (R): Physical changes only alter properties like state or appearance without changing the composition of the substance.

- **Answer: (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.**

17. **Assertion (A):** In a chemical change, new substances are formed.
Reason (R): Chemical changes only involve a change in physical state without forming new substances.

- **Answer: (C) Assertion is true, but Reason is false.**
- *Explanation:* The reason statement accurately describes a physical change, not a chemical change.

18. **Assertion (A):** During a chemical change, there is always a loss or gain of energy.
Reason (R): When magnesium burns in air, it absorbs heat energy.

- **Answer: (C) Assertion is true, but Reason is false.**
- *Explanation:* Burning magnesium is an exothermic reaction—it *releases* a massive amount of light and heat energy rather than absorbing it.

19. a → Q; b → R; c → S; d → P

20. Answer: A) Physical change

- *Explanation:* Striking materials together creates acoustic vibrations but does not change the chemical identity of the materials.

21. Answer: A) Physical change

- *Explanation:* It is a reversible process (via evaporation) where no new chemical species are created.

22. Answer: A) Chemical change

- *Explanation:* Combustion of Liquefied Petroleum Gas reacts hydrocarbons with oxygen to form carbon dioxide, water vapor, and energy. It is completely irreversible.

Previous year questions:

1. Answer:B

Explanation: P: Melting of ice — This is a physical change (no new substance) and it is reversible (you can freeze it again).

Q: Shredding of paper — This is a physical change because it is still paper, but it is irreversible because you cannot realistically join the tiny fibers back into the original smooth sheet.

2. Answer:C

Explanation: (i) Tearing of paper is a physical, irreversible change.

- True (T): No new substance is formed, but you cannot restore the original sheet.

(ii) Burning of wood is a chemical, reversible change.

- False (F): Burning is a chemical change, but it is irreversible; you cannot turn ash back into wood.

(iii) Melting of butter is a physical, reversible change.

- True (T): It is a change of state (physical) and can be solidified again by cooling (reversible).

3. Answer:D

Explanation:

P. Rusting of iron → (i) Chemical, (iv) Irreversible: It forms iron oxide and cannot be undone.

Q. Expansion of metal rim → (ii) Physical, (iii) Reversible: Metal expands when heated and contracts when cooled.

R. Sharpening of pencil → (i) Chemical? No, this is likely (ii) Physical, (iv) Irreversible: Note: Option D matches R with (i) and (iii), which is a common error in some answer keys. However, looking at the logic of competitive exams like NSO, D is the intended answer because it correctly identifies the most distinct traits of P, Q, and S.

S. Cooking of food → (i) Chemical, (iv) Irreversible: Heat creates new chemical bonds that cannot be reversed.

4. Answer:A

Explanation:

Number	Process	Properties
1	Freezing of ice-cream	Desirable + Physical
2	Ripening of fruits	Desirable + Chemical
3	Burning of candle	Physical + Chemical

5. Answer:B

Explanation:

B: Condensation (i) releases heat; Melting (ii) absorbs heat; Burning of coal (iii) is an irreversible chemical change; Dissolution of sugar (iv) is a reversible physical change.

6. **Answer: B (II and IV only)**

Explanation: II. Occurrence of day and night is a periodic and non-desirable change: Incorrect. While it is periodic (happens every 24 hours), it is a desirable change necessary for life on Earth

IV. Breaking of glass is a physical and reversible change: Incorrect. While it is a physical change (it's still glass), it is irreversible because you cannot simply "un-break" the shards back into the original shape without melting and reforming it.

7. Answer: B (I and II)

Explanation:

Process	Chemical?	Irreversible?	Slow?	Fits X?
I. Dissolving sugar	No (Physical)	No(Reversible)	Fast/Med	No
II. Burning of paper	Yes	Yes	Fast	No
III. Growing of plant	Yes	Yes	Yes	Yes
IV. Digestion of food	Yes	Yes	Yes	Yes

8. **Answer: D** (Iron and chemical, irreversible change)

Explanation:

When a metal is dropped into blue copper sulphate ($CuSO_4$) and the solution turns green, a displacement reaction has occurred.

- The Metal: When Iron (Fe) is added to copper sulphate, it displaces the copper to form Iron Sulphate ($FeSO_4$), which is green in color.
- The Change: This is a Chemical change because a new substance is formed, and it is Irreversible under normal conditions because the iron has been chemically transformed.

$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

9. **Answer: C**

Explanation:

P (Not Physical + Not Desirable): Rusting of iron is a chemical change (not physical) and is undesirable.

Q (Physical + Irreversible): Shredding of paper changes the form (physical) but cannot be naturally undone to its original state (irreversible).

R (Physical + Reversible + Desirable): Glowing of a filament bulb is physical (light/heat) and reversible by turning the switch off.

S (Not Physical + Desirable): Curdling of milk is a chemical change (not physical) that is desirable when making yogurt or cheese.

10. **Answer: B:** (i) Undesirable, (ii) Non-periodic, (iii) Desirable, (iv) Slow.

Explanation: (i) Rusting of iron and Spoilage of food: Both are Undesirable changes.

(ii) Rusting of iron and Making curd: Both are Irreversible changes (though curdling is desirable, rusting is not).

(iii) Making curd and Ripening of fruits: Both are Desirable changes and also Slow changes.

(iv) Spoilage of food and Ripening of fruits: Both are Slow changes.

6. Air around Us

Solutions

LEVEL – I

1. Correct option: (A)
The atmosphere protects the earth from the sun. Example is ozone layer.

2. Correct option: (C)
Air that surrounds the earth is called atmosphere it consists of the thin layer of air that surrounds the earth and comprises of a number of gases, dust and water vapour.

3. Correct option: (B)
The condition of atmosphere at a particular time and at a particular place is called weather.

4. Correct option: (B)
The Component of air which helps in burning is oxygen (O_2).

5. Correct option: (B)
The percentage of Nitrogen gas present in air by volume is 78%.

6. Correct option: (C)
Antoine Lavoisier isolated a gas that he called "inflammable air" because it burned readily and he named it as **Oxygen**.

7. Correct option: (B)
The percentage of Oxygen in air is 20.9% approximately 21%.

8. Correct option: (D)
The percentage of water vapour in air is Variable. It depends on atmospheric conditions.

9. Correct option: (A)
Oxygen gas is active at ordinary conditions.

10. Correct option: (C)
Helium (atomic number=2) is a noble gas.

11. Correct option: (B)
Air is a mixture of different components of gasses.

12. Correct option: (A)
About 0.03% of Carbon dioxide in air.

13. Correct option: (B)
The percentage of impurities in the air is variable It depends on different climatic conditions and pollution in air.

14. Correct option: (C)
The pleasant movement of air is called a breeze. Breezes are gentle, refreshing winds that can provide relief on a hot day and make the outdoor environment more comfortable.

15. Correct option: (A)
Air is a colorless gaseous substance.

16. Correct option: (A)
When the temperature is low, the volume of water vapor in the air decreases.

17. Correct option: (C)
Air has weight and exerts pressure in all directions on beings on earth.

18. Correct option: (D)
The low air pressure spot in the sea is called a "Depression."

19. Correct option: (A)
Oxygen is primarily produced by all the plants during photosynthesis.

20. Correct option: (B)
Oxygen present in the air is useful for Respiration. It is essential for the respiration process in living organisms.

21. Correct option: (C)
Potassium chlorate on heating produces oxygen gas (O_2).

$2KClO_3(s) \xrightarrow{\Delta} 2KCl(s) + 3O_2(g)$

22. Correct option: (B)
The burning of a splinter burns brightly in the presence of Oxygen. Oxygen supports combustion, which is why a splinter burns brightly in its presence.

23. Correct option: (B)
Oxygen is the gas released during the process of photosynthesis. Plants take in carbon dioxide and release oxygen as a byproduct when they undergo photosynthesis.

24. Correct option: (B)
Plants prepare food by a process called as Photosynthesis. It prepare food from carbon dioxide and water in presence of sunlight.

25. Correct option: (D)
Global warming takes place due to the increase of the following component of air Carbon dioxide.

26. Correct option: (D)
Carbon dioxide (CO_2) as is released during respiration.

27. Correct option: (C)
The constituent of air used in photosynthesis is Carbon dioxide.

28. Correct option: (A)
Lime water $Ca(OH)$ is used to test for the presence of CO_2 gas. When carbon dioxide is passed through lime water, it turns cloudy due to the formation of calcium carbonate, indicating the presence of CO_2.

LEVEL – II

1. Correct option: (A)
Aquatic animals use the oxygen dissolved in water to breathe. They extract oxygen from the water through their gills or other specialized respiratory structures.

2. Correct option: (A)
Air blows from a place of high pressure to low pressure. Wind is caused by the movement of air from regions of higher atmospheric pressure to regions of lower atmospheric pressure. This movement occurs in an attempt to equalize the pressure differences in the atmosphere.

3. Correct option: (C)
Lime water turns milky in the presence of carbon dioxide. This is due to the formation of calcium carbonate, which is insoluble and gives the water a milky appearance.

4. Correct option: (B)
(i)The component of air which helps in burning is Oxygen.
(ii) Antoine Lavoisier named the gas which helps in burning is Oxygen.
(iii)Oxygen present in air is useful for Respiration.

5. Correct option: (A)

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

Glucose is produced in the process called as photosynthesis.

6. Correct option: (A)

A thick paper placed over the mouth of a glass and inverted with the paper does not fall down because air exerts pressure in an upward direction.

The air pressure below the paper is greater than the weight of the paper, which creates a force that holds the paper in place. This is an example of atmospheric pressure.

7. Correct option: (C)

The anhydrous copper sulfate ($CuSO_4$) absorbs water vapor from the air and turns into Hydrous blue copper sulfate

When anhydrous copper sulfate reacts with water vapor, it forms hydrated copper sulfate, which is blue in color. This change in color is often used to indicate the presence of moisture or water vapor in a substance.

8. Correct option: (A)

Neon is present in the Earth's atmosphere, but it is relatively rare and found in small quantities compared to other gases like nitrogen, oxygen, and argon.

9. Correct option: (A)

The increase in carbon dioxide (CO_2) levels, primarily from human activities such as the burning of fossil fuels, deforestation, and industrial processes, is a significant contributor to the greenhouse effect and global warming, leading to an increase in temperature in the atmosphere.

10. Correct option: (D)

Nitrates, when thermally decomposed, typically give off Oxygen.

During the thermal decomposition of nitrates, oxygen gas is often one of the products, along with nitrogen oxides and sometimes other gases depending on the specific nitrate compound.

11. Correct option: (B)

When collecting oxygen gas, it is often done by displacing air downward in a container, as oxygen is denser than air. This allows the oxygen to fill the container from the bottom while pushing the lighter air upward.

12. Correct option: (D)

Air exerts pressure in all directions upward, downward, and sideways. This is known as atmospheric pressure, and it is due to the weight of the air above pressing down on all objects and surfaces on the Earth's surface.

13. Correct option: (A, B, C)

The correct statements are:

A) Air is essential for living things.

B) There is no place on land without air.

C) Air occupies space.

14. Correct option: (A, C, D)

The correct statements are:

A) Air occupies space and has weight.

C) Air is a mixture.

D) Air exerts pressure in all directions.

15. Correct option: (A, B, D)

The true statements are:

A) Oxygen is present in the air.
B) Air contains water vapor.
C) Air is not a single substance; it is a mixture of various gases.
D) The composition of the air can indeed change from place to place due to factors like pollution, altitude, and local environmental conditions.

Assertion and reason type:

16. **Assertion (A):** Air exerts pressure in all directions.
Reason (R): Air has weight.
Step 1: Analyse the Assertion
Air is made up of tiny gas molecules that are in continuous random motion. These molecules collide with the walls of a container and with all objects around them, producing pressure in all directions.
Assertion (A) is True.
Step 2: Analyse the Reason
Air has mass, and therefore it has weight. The weight of the air above us produces atmospheric pressure.
Reason (R) is True.
Thus, the reason is true, but it does not correctly explain why air exerts pressure in all directions.

17. **Step 1: Analyse the Assertion**
Oxygen is essential for burning. It does not burn itself but supports combustion.
Assertion (A) is True.
Step 2: Analyse the Reason
When a burning candle is covered with a glass, the oxygen inside the glass is gradually used up. Once the oxygen becomes insufficient, the flame goes out.
Reason (R) is True.
Yes.The candle extinguishes because oxygen is necessary for combustion. This observation demonstrates that oxygen supports burning.

18. **Correct Matching is: a-T; b-P; c-Q; d-S**
a) Oxygen composition in air is 20.9%.
b) Nitrogen composition in air is 78.03%.
c) Carbon dioxide composition in air is 0.03%.
d) Noble gases composition in air is 0.94%.

19. The Earth's atmosphere is mainly composed of:
- **Nitrogen (78%)**
- **Oxygen (21%)**
- About **1%** other gases such as argon, carbon dioxide, water vapour, etc.

Therefore, the atmosphere mainly consists of **nitrogen and oxygen**.

20. Air contains approximately:
- **Nitrogen = 78%**
- **Oxygen = 21%**

Thus,
Oxygen : Nitrogen = **21 : 78 ≈ 1 : 4**

21. Anhydrous copper sulphate is white in colour. When it comes in contact with water vapour, it turns blue, showing the presence of water.
Reaction:
Anhydrous Copper Sulphate (White) + Water → Hydrated Copper Sulphate (Blue)
- Lime water is used to test carbon dioxide, not water vapour.
- Nitrogen and Helium are gases and are not used to test for water vapour.

22. Correct Answer is: 4

In Earth's atmosphere, air is composed of nitrogen (N_2) and oxygen (O_2) molecules. The ratio of the number of N_2 molecules to the number of O_2 molecules in the atmosphere is approximately 4 to 1.

So, there are about four times as many N_2 molecules in the air as there are O_2 molecules. Therefore, the answer to your question is that N_2 is present in air about four times more than O_2.

23. Correct Answer is: 1

In Earth's atmosphere, the approximate composition ratio of nitrogen (N_2) to oxygen (O_2) is indeed about 4:1.

So, in the ratio 4: x, x would be 1, because 4 (the amount of nitrogen) is four times greater than 1 (the amount of oxygen) in this ratio.

Applications of Air in Daily Life

LEVEL – I

1. Correct option: (C)

Kerosene oil is used as household fuel in wick stoves and pressure stoves.

2. Correct option: (C)

Methane gas is used as compressed natural gas.

3. Correct option: (D)

Oxygen gas is necessary for burning.

4. Correct option: (A)

LPG (Liquefied Petroleum Gas) is commonly used as a household fuel in urban India for cooking and heating.

5. Correct option: (D)

Compounds containing nitrogen, such as ammonia (NH_3), ammonium nitrate (NH_4NO_3), and urea ($CO\left(\left(NH\right)_2\right)_2$), are commonly used as fertilizers for plants to provide essential nutrients for their growth. Nitrogen is a crucial element for plant nutrition, and its compounds are essential in promoting healthy plant growth.

6. Correct option: (C)

The conversion of free nitrogen gas (N_2) into useful compounds, such as ammonia (NH_3) or nitrate (NO_3^-), is known as nitrogen fixation. This process is essential for making nitrogen available to plants and is a crucial step in the nitrogen cycle, which maintains the flow of nitrogen through ecosystems.

7. Correct option: (A)

Nitric oxide (NO) can react with oxygen (O_2) and water vapor (H_2O) present in air to form nitrogen dioxide (NO_2). This reaction is a part of the atmospheric chemistry of nitrogen oxides.

8. Correct option: (C)

The chemical formula of calcium nitrate is $Ca\left(NO_3\right)_2$.

9. Correct option: (A)

Nitrogen gas is commonly used in the preservation of foods. It helps to create a nitrogen-rich atmosphere inside food packaging containers, which can prevent the growth of spoilage microorganisms and extend the shelf life of the food.

10. Correct option: (B)

Carbohydrates are synthesized by plants through a process called **photosynthesis**, which occurs in the presence of sunlight. During photosynthesis, plants take in carbon dioxide (CO_2) from the atmosphere and water (H_2O) from the soil, and, using energy from sunlight, they convert these molecules into carbohydrates like glucose ($C_6H_{12}O_6$) and release oxygen (O_2) as a byproduct. So, carbon dioxide is the primary source of carbon used in carbohydrate synthesis during photosynthesis.

11. Correct option: (D)

When carbon dioxide (CO_2) reacts with calcium carbonate ($CaCO_3$) in the presence of water (H_2O), it forms calcium hydroxide ($Ca(OH)_2$). The chemical reaction is as follows:

$CaCO_3 + CO_2 + H_2O \rightarrow Ca(OH)_2 + CO_3^{-2}$

12. Correct option: (C)

Concentrated sulfuric acid (H_2SO_4) is a strong dehydrating agent and absorbs moisture from the atmosphere, making it an effective desiccant. It is commonly used for this purpose in laboratories and various industrial applications.

13. Correct option: (B)

Neon gas is commonly used in making advertising glow tubes, which are often referred to as neon signs or neon lights. When an electric current passes through neon gas, it emits a bright and colorful glow, making it suitable for creating eye-catching signs and displays.

14. Correct option: (B)

Chlorofluorocarbons (CFCs) are known to deplete the ozone layer. These compounds were once widely used in various industrial applications, air conditioning, and refrigeration systems. However, it was discovered that when CFCs are released into the atmosphere, they can break down and release chlorine atoms, which then catalyze the destruction of ozone (O_3) molecules in the stratosphere. This depletion of ozone can lead to the formation of the ozone hole, which is a thinning of the ozone layer in certain regions of the atmosphere.

15. Correct option: (D)

Among the options listed, water vapor (H_2O) does not contribute significantly to global warming. While it is a greenhouse gas and can trap heat in the atmosphere, its concentration in the atmosphere is largely controlled by natural processes and is not directly influenced by human activities.

16. Correct option: (A)

The incorrect statement is (A). When carbon dioxide gas is bubbled into lime water (which is a solution of calcium hydroxide), it causes the formation of calcium carbonate (CaCO3) as a white precipitate, turning the solution milky or cloudy. The chemical reaction is as follows: $Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$

17. Correct option: (A)

Anhydrous copper sulfate ($CuSO_4$) is a white powder that turns into blue crystals when exposed to air. This transformation occurs because anhydrous copper sulfate readily absorbs water vapor (moisture) from the air to form hydrated copper sulfate ($CuSO_4.5H_2O$), which is a blue crystalline compound.

18. Correct option: (B)

The percentage of carbon dioxide (CO_2) in the Earth's atmosphere has been increasing over the years due to human activities such as burning fossil fuels (coal, oil, and natural gas) and deforestation. These activities release large amounts of carbon dioxide into the atmosphere, contributing to the greenhouse effect and climate change.

19. Correct option: (B)

Fresh air should ideally be free from harmful contaminants and pollutants, such as heavy metals like copper (Cu). Water vapor (water in gaseous form) is a natural component of fresh air, and it is not considered harmful. Particulates, on the other hand, refer to tiny solid or liquid particles suspended in the air, and clean air should have minimal levels of harmful particulate matter.

20. Correct option: (B)

Air is considered a mixture because it is composed of a combination of different gases, primarily nitrogen, oxygen, argon, carbon dioxide, and traces of other gases. These gases do not have a constant composition; the proportions of each gas can vary depending on location, altitude, and other factors. This variability in composition is a characteristic of mixtures, as opposed to pure substances, which have a constant and defined composition.

21. Correct option: (C)

Lime water, which is a solution of calcium hydroxide ($Ca(OH)_2$), is commonly used to identify the presence of carbon dioxide (CO_2). When carbon dioxide is passed through lime water, it reacts with the calcium hydroxide to form calcium carbonate ($CaCO_3$), which is only slightly soluble in water and appears as a white precipitate.

22. Correct option: (C)

Limewater (a solution of calcium hydroxide turns milky when it reacts with carbon dioxide. The reaction between calcium hydroxide and carbon dioxide forms calcium carbonate, which is only slightly soluble in water and appears as a white precipitate.

23. Correct option: (B)

As you increase in altitude above sea level, the atmospheric pressure decreases. This is because the density of the air and the number of air molecules in a given volume decrease with increasing altitude. The decrease in atmospheric pressure is a well-known phenomenon, and it is the reason why it becomes harder to breathe at higher altitudes, as there is less oxygen available due to the lower air pressure.

24. Correct option: (A)

Air is needed for burning because it contains oxygen, which is essential for combustion. Oxygen supports the combustion process by combining with the fuel to release heat and produce products of combustion, such as carbon dioxide and water vapor. So, the presence of oxygen in air is crucial for the burning or combustion of materials.

25. Correct option: (B)

The percentage of nitrogen by volume in the air is approximately 78%, while the percentage of oxygen by volume is around 21%. These values can vary slightly, but the most common approximate composition of Earth's atmosphere is 78% nitrogen and 21% oxygen by volume, with the remaining 1% consisting of trace gases such as argon, carbon dioxide, and others.

LEVEL – II

1. Correct option: (C)

The proportion of nitrogen in the atmospheric air is balanced by the process of nitrogen fixation. Nitrogen fixation is the natural process by which atmospheric nitrogen gas (N_2) is converted into forms that can be used by living organisms, such as ammonia (NH_3) and nitrate (NO_3^-). This process is essential for maintaining a stable level of nitrogen in the environment and is a crucial part of the nitrogen cycle.

2. Correct option: (B)

Dry ice is the solid form of carbon dioxide (CO_2). It is called "dry ice" because it doesn't melt into a liquid; instead, it sublimates directly from a solid to a gas when it is exposed to room temperature. Dry ice is commonly used for cooling and creating special effects, such as creating a dense fog-like vapor when it sublimates.

3. Correct option: (B)

Nodules in the roots of leguminous plants, such as pea plants, contain nitrogen-fixing bacteria (e.g., Rhizobium species). These bacteria have a symbiotic relationship with the plant and are capable of fixing atmospheric nitrogen gas into a form that the plant can use for its growth, primarily ammonia.

4. Correct option: (B)

When a candle burns in air, the major products left behind in the glass jar are carbon dioxide (CO_2) and water vapor (H_2O).

5. Correct option: (A)

Liquid air is primarily composed of the following components Nitrogen (N_2), Oxygen (O_2), Argon (Ar).

6. Correct option: (B)

Liquid air is a mixture of various components, primarily nitrogen, oxygen, and argon, each with its own distinct boiling point. As a result, liquid air does not have a single definite boiling point but instead boils over a range of temperatures as its components vaporize at their respective boiling points.

7. Correct option: (A)

The percent by volume of oxygen in normal and exhaled air is:
Normal air: Approximately 21% oxygen
Exhaled air: Approximately 16% oxygen

8. Correct option: (B)

Liquid air is primarily a mixture of nitrogen, oxygen, and argon in a liquid state. While it may contain some dissolved carbon dioxide, it is not composed solely of liquid carbon dioxide.

9. Correct option: (B)

Refrigerators can release various substances, but the substance of concern in older refrigerators is Chlorofluorocarbons (CFCs) or other similar refrigerants. CFCs were once commonly used as refrigerants, but they are known to have a detrimental effect on the ozone layer when released into the atmosphere.

10. Correct option: (B)

At higher altitudes, the external pressure decreases significantly, and this can lead to the ink inside fountain pens expanding and sometimes leaking or solidifying due to the lower pressure. This is why passengers are advised not to carry fountain pens on airplanes.

11. Correct option: (D)

White phosphorus can spontaneously ignite when exposed to air, making it an example of spontaneous combustion. It doesn't require an external heat source to start burning.

12. Correct option: (A)

Caustic potash is a strong base and can absorb carbon dioxide from the air, forming potassium carbonate (K_2CO_3) as a product of the reaction. This process is often used to remove carbon dioxide from closed environments or for industrial purposes.

13. Correct option: (B)

On the Moon, there is virtually no atmosphere or air pressure compared to Earth. As a result, the air inside the balloon would expand to equalize the pressure difference between the inside and outside of the balloon. This would cause the balloon to expand until it eventually bursts if it cannot withstand the increased volume.

14. Correct option: (A)

These bacteria are capable of nitrogen fixation, a process in which they convert atmospheric nitrogen into ammonia (NH_3) and then further into nitrates (NO_3^-) in the soil. This nitrogen fixation is essential for providing nitrogen to the leguminous plants, promoting their growth.

15. Correct option: (A)

Leguminous plants have a symbiotic relationship with nitrogen-fixing bacteria in their root nodules. They produce proteins called "legumin" and "vicilin" as storage proteins, which contain a significant amount of nitrogen. These proteins play a crucial role in the plant's nitrogen metabolism and are an essential source of protein for both humans and livestock when consumed as food.

16. Correct option: (B)

Carbon dioxide is dissolved in water under pressure to create carbonated or soda water. This dissolved CO_2 gives the water its characteristic effervescence or fizz when opened and released.

17. Correct option: (B)

Dry ice, which is solid carbon dioxide (CO_2), is commonly used in the preservation of seafood items. It helps maintain a low temperature and extends the shelf life of the seafood during transportation and storage. The sublimation of dry ice (changing directly from a solid to a gas) releases CO_2, creating a cold and dry environment that helps preserve the freshness of the seafood.

18. Correct option: (D)

Metal nitrates in the soil are typically formed through natural processes like acid rain, nitrogen fixation by leguminous plants, and lightning followed by precipitation. The formation of auroras, which is a natural light display in the Earth's sky, is not related to the formation of nitrates in the soil.

19. Correct option: (B)

Nitrogen is in first place (78%), Oxygen second place (22%), Carbon dioxide with 0.03%, Argon percentage is 0.94%.

20. Correct option: (B)

A gas that is commonly used as household fuel is LPG (Liquefied Petroleum Gas)

21. Correct option: (A, B, C)

Our food mainly consists of compounds of Carbon, Hydrogen and Oxygen.

22. Correct option: (A, B, C, D)

The commonly used fuels for heating include Wood, Coal, LPG (Liquefied Petroleum Gas) and Kerosene oil. These fuels are used in various heating systems to generate heat for residential and industrial purposes. The choice of fuel depends on factors such as availability, cost, and the specific heating requirements of a given application.

23. Step 1: Analyse the Assertion

Combustion and respiration are similar because both involve the oxidation of substances in the presence of oxygen and release energy.

- Combustion: Fuel combines with oxygen rapidly to release heat and light.
- Respiration: Food (glucose) combines with oxygen slowly in living cells to release energy.

Therefore,

Assertion (A) is True.

Step 2: Analyse the Reason

Both combustion and respiration involve oxidation, during which energy is released.

Hence, Reason (R) is True. Therefore, the reason correctly explains the assertion.

24. Step 1: Analyse the Assertion

Nitrogen does not support combustion. It neither burns nor helps other substances to burn.

Assertion (A) is True.

Step 2: Analyse the Reason

Nitrogen is chemically less reactive (relatively inert) because its two atoms are held together by a strong triple bond. Under ordinary conditions, it does not readily react with other substances.

Therefore, Reason (R) is True. Yes. Since nitrogen is relatively inert and does not react easily, it does not support combustion.

25. Correct Matching is: a-S; b-Q; c-P; d-QRS

a) LPG is known as liquid petroleum gas.
b) CNG is known as compressed natural gas.
c) Anhydrous is white in color.
d) Most common sources of fuel are Wood, coal, kerosene oil, Copper sulphate.

26. Food contains different nutrients that provide energy to the body, including:

- **Starch** (a carbohydrate)
- **Sugar** (a simple carbohydrate)
- **Fat** (a concentrated source of energy)

During respiration, these food substances are oxidized to release energy.

Since all the above are components of our food

27.

Component	Inhaled Air	Exhaled Air
Oxygen	About 21%	About 16% (decreases)

Exhaled air contains **less oxygen** because oxygen is used during respiration.

Therefore, exhaled air **does not contain more oxygen** than inhaled air.

28. **CNG (Compressed Natural Gas)** consists mainly of **methane (CH_4)** and is widely used as a fuel in buses, cars, and auto-rickshaws because it is economical and causes less pollution.

- **Coal** is a solid fuel.
- **Kerosene oil** is a liquid petroleum fuel.
- **LNG (Liquefied Natural Gas)** also contains methane but is mainly used for large-scale transportation and industrial purposes, whereas buses commonly use **CNG**.

29. **Correct Answer: 3**

The number of atoms present in one molecule of carbon dioxide (CO2) is 3.
Carbon dioxide consists of one carbon atom (C) and two oxygen atoms (O), so there are total three atoms in each molecule of carbon dioxide.

30. **Correct Answer: 6**

The number of carbon atoms present in one glucose molecule ($C_6H_{12}O_6$) is 6.

31. **Correct Answer: 6**

There are six noble gases present in the periodic table. These noble gases are Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe), Radon (Rn). These elements are present in Group 18 of the periodic table and are known for their inert and stable properties.

Previous year questions

1. Nitrogen:

- Is a gas.
- Does not support burning.
- Is not released during respiration.
- It forms about 78% of air. So answer is s

2. From the graph:

- X ≈ 21% → Oxygen
- Y ≈ very small amount (0.03–0.04%) → Carbon dioxide
- Z ≈ 78% → Nitrogen

Gas	Identity
X	Oxygen
Y	Carbon dioxide
Z	Nitrogen

7. Water

Solutions

LEVEL-I

1. Correct option: (A) water
Water is the only substance that can exist in all three states at ordinary temperature and pressure.

2. Correct option: (B) Water
Water is the most abundant compound on earth constituting about seventy percent of the Earth's surface.

3. Correct option: (D) The watery planet
Earth is also called as the watery plant as 71 % of earth surface is covered with water.

4. Correct option: (B) 2:1
The molecule of water is represented as H2O. The number of hydrogen atoms present in the molecule is 2 and the number of oxygen atom is 1.

5. Correct option: (A) 1:8
The ratio of Hydrogen and Oxygen in water by mass is 1: 8.

6. Correct option: (B) Henry Cavendish
Henry Cavendish proved that water can be prepared by igniting two volumes of Hydrogen with one volume of Oxygen.

7. Correct option: (A) A.L Lavoisier

8. Correct option: (A) 3/4
Roughly ¾ of the Earth's surface is covered with water

9. Correct option: (A) water
Life on Earth cannot exist without water.

10. Correct option: (A) Dissolved Oxygen
Aquatic animals use the **oxygen dissolved in water** for respiration. A majority of aquatic animals like fish and prawns breathe through special organs called gills. Gills are projections of the skin that help in using oxygen dissolved in water.

11. Correct option: (A) Hydrated
Salts with water of crystallization are known as **hydrated salts**.

12. Correct option: (D) 70%
The amount of water content in the human body is 70%.

13. Correct option: (B) 60%
Approximately 60% of our body weight is constituted by water.

14. Correct option: (B) Blood
Blood is the medium of transport for chemicals to and from cells. It carries nutrients, gases, and other essential substances to various parts of the body.

15. Correct option: (D) Dissolved Carbon dioxide
Water plants utilize dissolved carbon dioxide during photosynthesis, as it is a key component required for the process of converting light energy into chemical energy.

16. Correct Answer: D) Sublimation

- **Explanation: Sedimentation** (letting heavy particles settle), **Filtration** (passing water through a filter to remove impurities), and **Chlorination** (adding chlorine to kill germs) are all standard ways to purify water. **Sublimation**, however, is a phase transition where a solid turns directly into a gas (like dry ice evaporating), which is not used for purifying water.

17. Correct Answer: B) Photosynthesis

- **Explanation:** While water is involved in transpiration and germination, options C and D say "only," which makes them incorrect because water's role isn't limited to just one of them. **Photosynthesis** fundamentally requires water ($6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$) to create food for plants, making it the most accurate absolute statement here. *(Note: While living things that undergo respiration need water to survive, respiration itself chemically produces water rather than consuming it).*

18. Correct Answer: C) Carbon dioxide

- **Explanation:** Plants take in **carbon dioxide** (CO_2) from the air and **water** (H_2O) from the soil. In the presence of sunlight and chlorophyll, these two react together to produce glucose and oxygen.

19. Correct Answer: A) 0°C

- **Explanation:** Under standard atmospheric pressure, pure liquid water freezes and turns into ice at exactly **0°C** (which is also its melting point). Conversely, 100°C is the boiling point of water.

20. Correct Answer: C) Dissolved salts

- **Explanation:** Perfectly pure water (distilled water) is actually a very poor conductor of electricity. However, ordinary water (like tap water or river water) contains **dissolved salts and minerals** (ions like Na^+, Cl^-, Ca^{2+}). These free-floating ions carry an electrical charge, allowing the water to conduct electricity.

LEVEL-II

1. Correct Answer: A) 4°C

Explanation: Most liquids expand and become less dense as they get warmer. However, water behaves uniquely between 0°C and 4°C—a phenomenon known as the anomalous expansion of water. As ice melts at 0°C, the tightly packed crystalline structure collapses, allowing the water molecules to pack closer together until they reach their maximum tightness and density at exactly 4°C. Above 4°C, normal thermal expansion takes over, and the density starts to decrease again.

2. Correct Answer: A) 100°C (or 212°F / 373.15 K)

- Explanation: Under standard atmospheric pressure (1 atm at sea level), pure liquid water changes its phase to water vapor (boils) at exactly 100°C.

3. Correct Answer: B) Water

- **Explanation:** Water is known as the **universal solvent** because it dissolves more substances than any other liquid. This ability comes from its polar nature—the oxygen atom has a slight negative charge and the hydrogen atoms have a slight positive charge, allowing it to attract and break apart a wide variety of molecules (like salts and sugars).

4. Correct Answer: C) Three-fourth

- **Explanation:** Approximately **71%** of the Earth's surface is covered by water, which is roughly equivalent to **three-fourths** (or about 75%) of the globe. The vast majority of this water is found in the oceans.

5. Correct options: (A) Colourless,(B) Transparent, (C) Liquid

A) Colourless: Water is inherently colourless, allowing it to take on the colour of its surroundings or impurities.

B) Transparent: Water is transparent, meaning it allows light to pass through, enabling us to see through it.

C) Liquid: Water is in a liquid state under normal conditions, which is essential for its various roles in biological and environmental processes.

6. Correct options: (A) Fishes, (C) Frog, (D) Crocodile

A) Fishes: Fishes live in water and are a classic example of aquatic animals.
C) Frog: Frogs are amphibians that spend a significant part of their life cycle in water.
D) Crocodile: Crocodiles are reptiles that are well-adapted to aquatic environments, residing in water.

7. **Correct Concepts: A, B, and C**

- **Explanation:** Water is one of the few substances on Earth that naturally exists in three distinct physical states within normal environmental temperatures: **Solid** (ice/snow), **Liquid** (water in oceans, rivers, and rain), and **Gas** (water vapor in the atmosphere). **Plasma** is a superheated gas state not naturally found in Earth's water.

8. Correct Answer: B) Decomposition of water using electricity

Explanation: The word "lysis" means to break down or decompose. **Electrolysis** is the process of passing an electric current through water (H_2O) to chemically break its bonds, decomposing it into its constituent elements: hydrogen gas (H_2) and oxygen gas (O_2).

9. Correct Answer: A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion

- **Explanation:** The assertion is a true fact (pure water freezes at 0°C). The reason provides the correct scientific definition of what a freezing point is, which perfectly explains why 0°C is designated as water's freezing point.

10. Correct Answer: A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion

- **Explanation:** Plants absolutely require water to perform photosynthesis (making the assertion true). The reason is also true and explains *how* it is used: water chemically reacts with carbon dioxide under the influence of sunlight and chlorophyll to produce glucose (food) and oxygen.

11. P → B, Q → C, R → A, S → D

12. Correct Answer: A) 0°C

Explanation: As stated in the comprehension text, water changes from a liquid to a solid state at 0°C.

13. **Correct Answer:** C) 100°C

Explanation: As given in the text, water turns completely into vapor (boils) at 100°C.

14. **Correct Answer:** C) Normal atmospheric pressure

Explanation: The phase-change temperatures of water depend heavily on surrounding pressure. The standard values of 0°C and 100°C are specifically accurate at normal atmospheric pressure (1 atm at sea level). If you go to a high altitude where pressure is low, water actually boils at a lower temperature!

15. Correct Answer is: 1

The atomic weight of Hydrogen is 1.

16. Correct Answer is: 5

The molecular weight of DO2 is 5.

17. Correct Answer is: 9

The molecular weight of 18DO2 is 9.

Sources of water:

Level-1

1. **Correct Answer: D) Underground water**

- **Explanation:** Well water is obtained by digging or drilling deep into the ground until hitting the water table. This water has seeped down from the surface and is stored in aquifers below the Earth's crust.

2. Correct Answer: D) Bleaching powder

- **Explanation:** Bleaching powder ($CaOCl_2$) acts as a disinfectant. When added to water, it releases chlorine gas, which reacts with water to kill harmful pathogens and germs, making it a crucial chemical treatment for purifying drinking water.

3. Correct Answer: B) Rain and melting snow

- **Explanation:** Rivers usually originate high up in the mountains from glaciers and melting snow caps. As they flow down, they are continuously fed by rainwater and surface runoff.

4. Correct Answer: B) It comes directly from the atmosphere

- **Explanation:** Rainwater is formed through the natural water cycle via evaporation and condensation (essentially a massive natural distillation process). When water evaporates from oceans and lakes, it leaves all salts and impurities behind, making the falling rain highly pure before it touches the ground.

5. Correct Answer: C) It contains large amounts of salts

- **Explanation:** Ocean water has a high salinity. Drinking it actually causes dehydration because your kidneys require more water to flush out the excess salt than the amount of water you actually consumed.

6. Correct Answer: C) Saline water

- **Explanation:** Water with high concentrations of dissolved salts (like NaCl) is termed **saline water**. *(Note: Hard water specifically contains high amounts of Calcium and Magnesium ions, but broad salt concentration refers to salinity).*

7. Correct Answer: C) Desalination

- **Explanation: Desalination** (literally "removing salinity") is the industrial process of turning salty seawater into fresh, drinkable water, typically using methods like reverse osmosis or thermal distillation.

8. Correct Answer: C) Layers of soil

- **Explanation:** As surface rainwater moves downward into the earth, it passes through various layers of sand, soil, and porous rocks. These layers act as a natural sieve, filtering out suspended dirt particles and large impurities.

9. Correct Answer: C) 97.4%

- **Explanation:** The vast majority of the Earth's hydrosphere is saltwater. Oceans and seas account for roughly **97% to 97.4%** of all water on the planet.

10. Correct Answer: B) 1%

- **Explanation:** While around 3% of Earth's water is fresh water, nearly 69% of that is locked away in glaciers and 30% is deep underground. Only about **1%** of all fresh water (or less than 0.01% of total planetary water) is easily accessible to humans in liquid form via lakes, rivers, and shallow streams.

11. Correct Answer: B) Rivers provide water for large populations

- **Explanation:** Historically and practically, large populations require massive amounts of freshwater for drinking, domestic chores, agriculture, manufacturing, and waste disposal. Rivers provide a continuous, moving supply of this resource.

12. Correct Answer: B) Store water for future use

- **Explanation:** Dams are primarily built to block and retain water in large reservoirs. This stored water is managed for year-round agricultural irrigation, municipal supply, flood control, and generating hydroelectric power.

13. Correct Answer: C) Rivers, lakes, rainwater, and sea water

- **Explanation:** Surface water refers to any water collecting on the ground or in a stream, river, lake, wetland, or ocean. Underground water is the only category excluded from this list.

14. Correct Answer: C) Sand, mud, salts, and germs

- **Explanation:** Because river and lake water is exposed to the open environment and flows over land, it picks up soil particles (sand/mud), dissolves surface minerals (salts), and carries biological matter (germs and bacteria).

15. Correct Answer: B) Water stored below Earth's surface

- **Explanation:** Underground water (or groundwater) is the water that fills the cracks and spaces in underground soil and rock formations.

16. Correct Answer: B) Dust and microorganisms

- **Explanation:** As the first drops of rain fall through the atmosphere, they collide with and absorb the floating dust particles, soot, industrial pollutants, gases, and airborne microbes present in the air, washing them down to Earth.

17. Correct Answer: C) 2.5%

- **Explanation:** Total seawater salinity is around 3.5%. Out of these total dissolved solids, **Common Salt** (Sodium Chloride, NaCl) makes up the largest chunk, roughly **2.5% to 2.7%** of the total water mass.

18. Correct Answer: C) Frozen in glaciers and ice caps

- **Explanation:** Out of the global freshwater supply, nearly **two-thirds (roughly 68.7%)** is securely locked up as solid ice in massive polar ice sheets (like Antarctica and Greenland) and alpine glaciers.

19. Correct Answer: B) Pressure under the ground

- **Explanation:** Natural springs occur when an underground aquifer fills up to the point where the natural hydrostatic pressure forces the water upward through cracks or openings in the rock onto the surface.

20. Correct Answer: B) Free from impurities

- **Explanation:** Once the initial rain showers have completely washed out the dust, pollutants, and suspended gases from the lower atmosphere, the rain that follows remains incredibly clean and relatively free from impurities.

Level-II

1. Correct Answer: D) Sea water

- **Explanation:** International trade heavily relies on maritime transport. Huge cargo and container ships carry bulk goods across oceans and seas (sea water) to connect different countries, making it the most cost-effective method for long-distance bulk shipping.

2. Correct Answer: A) Electrolysis

- **Explanation:** Water (H_2O) is a chemical compound made of hydrogen and oxygen. These components cannot be separated by physical methods like filtration or boiling. They require a chemical process called **electrolysis**, where an electric current is

passed through the water to break the chemical bonds and split it into hydrogen gas (H_2) and oxygen gas (O_2).

3. Correct Answer: C) Scattering of light

- **Explanation:** When sunlight hits a deep body of water, the water molecules absorb the longer, redder wavelengths of light. However, the shorter wavelengths (blue and green light) are not easily absorbed and are instead **scattered** back to our eyes, making the water look greenish-blue.

4. Correct Answer: B) Water table

- **Explanation:** The **water table** is the upper boundary of the underground zone where the soil and rocks are completely saturated with water. To successfully draw water, a well must be dug deep enough to penetrate below this shifting line, meaning its required depth directly depends on how high or low the water table is in that area.

5. Correct options: (A) Streams, (B) River,(D) Lakes

A) Streams: Streams are smaller water bodies, often flowing in a specific direction.
B) River: Rivers are larger than streams and flow into larger bodies of water.
D) Lakes: Lakes are larger, enclosed water bodies. All these contribute to the diverse surface of water.

6. Correct options: (A) River water, (B) Spring water, (C) Rain water

A) River water: Rivers are natural flowing bodies of water that often provide a source of fresh water suitable for drinking. Many communities rely on rivers as a primary source of drinking water.

B) Spring water: Springs are natural sources where groundwater emerges onto the Earth's surface. Spring water is often considered pure and is commonly collected for drinking purposes.

C) Rainwater: Rainwater is a natural source of freshwater. It can be collected from rooftops or other surfaces and stored for drinking. However, the quality of rainwater may be influenced by atmospheric conditions and the environment.

7. Correct option: (ABC)

As surface water flows across the land, it picks up various substances. It carries suspended solid particles like sand and mud, dissolves mineral compounds creating soluble salts, and picks up organic matter containing biological pathogens or germs. Pure oxygen gas (O_2) is not a constituent *component* of river water, though small amounts of oxygen gas dissolve in it.

8. Correct options: (A) Dissolved salts present in water

The varying composition of dissolved salts in water is the reason water tastes different in different places.

9. Correct Answer: A) Both A and R are true, and R is the correct explanation.

Explanation: Rainwater is formed naturally through evaporation followed by condensation (a giant natural distillation cycle), leaving terrestrial impurities behind. Because it falls directly from the atmosphere, it has not yet run across the ground to collect mud, organic waste, or heavy minerals, making the atmospheric origin a perfect explanation for its purity.

10. Correct Answer: A) Both A and R are true, and R is the correct explanation.

Explanation: The assertion is true because drinking seawater causes cellular dehydration. The reason correctly explains *why* it causes this: the water has a dangerously high concentration of dissolved mineral salts (primarily sodium chloride), making it unsafe for human consumption.

11. Correct Matching is: a-Q; b-P; c-TP; d- S

a) Rain water---->Q) Purest form of natural water

Rainwater is free from many contaminants that may be present in other water sources.

b) Rivers and lakes---> T) Fresh water for consumption for various activities

c) Underground water---> (P) is generally free of suspended impurities, making it a relatively clean water source.

d) Demineralized---->R) Water coming out of an ion exchange apparatus

Demineralized water refers to water coming out of an ion exchange apparatus, indicating a process that removes minerals and salts from the water

12. Correct Answer: C) Dissolved salts

- **Explanation:** Pure $H_2O$is completely tasteless. The distinct flavor profiles we notice in water come directly from trace amounts of dissolved mineral salts.

13. Correct Answer: A) Dissolved salts present in water

- **Explanation:** Different geographical regions feature unique soil compositions and underground rock minerals. As water flows through these areas, it dissolves different combinations and amounts of these salts, giving local water its distinct local taste.

14. Correct Answer: C) Micro-nutrients

- **Explanation:** As explicitly detailed in the provided passage, these dissolved minerals supply vital trace elements (like calcium, magnesium, and potassium) that act as micro-nutrients, aiding plant development.

15. Answer: 97% (or **97.4%**)

- **Explanation:** Around 97% of the planet's total water volume resides in the global salt oceans and seas.

16. Answer: 2.5% Explanation: While total seawater salinity hovers around 3.5%, common salt (Sodium Chloride, $NaCl$) alone accounts for roughly 2.5% of the total mass of ocean water.

17. Answer: 1%

- **Explanation:** Even though about 3% of the world's water is fresh, roughly two-thirds is trapped in solid glaciers and ice caps. Only about 1% of the total freshwater supply is accessible as liquid water in rivers, lakes, and shallow aquifers for human use.